

QuaRK: Quantum Reservoir Kernels for Temporal Learning

Anonymous authors

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Abstract

QuaRK studies frozen quantum reservoir learning for causal temporal regression when data are dependent, measurements are finite, and only a finite pool of random reservoir realizations can be searched. The architecture exposes qubit width, spatial multiplexing, and memory range, while grouped Pauli measurements and an RMS-normalized, norm-controlled Matérn readout make the remaining statistical and measurement budgets explicit without letting feature dimension alone distort kernel geometry. A graph-local operator-spreading theorem shows that product input injection cannot enlarge Pauli support, whereas a generic cycle mixer activates higher-support components and therefore exposes higher-order projected-input interactions directly in measured features; this is a finite-resource representation statement, not a claim that entanglement creates new encoding frequencies or guarantees lower risk. We replace validation-based reservoir ranking by full-sample empirical-risk minimization over the finite frozen pool and give a uniform beta-mixing bound that controls the resulting data-dependent selection. A width-resolved approximation theorem separates information lost by the shared Gaussian projection from approximation by the reservoir family: under a canonical coupled projection, the resulting population information floor is nonincreasing with width and vanishes almost surely once the projection is injective, while finite spatial multiplexing and finite-norm Matérn readouts introduce no additional asymptotic floor. Continuity and full support then give positive probability to neighborhoods of good frozen designs, so a pool of size S misses all such designs with probability exactly $(1 - \mathbf{p}_{\text{hit}})^S$. Combining this pool-coverage term with dependent-data generalization, finite-shot error, and finite-window error yields an end-to-end excess-risk certificate for the predictor actually returned. Under a geometric beta-mixing envelope, a raw-time gap logarithmic in sample size is sufficient to keep the unit-block coupling term controlled. A separate Johnson–Lindenstrauss result is retained only as a finite-data geometry certificate, not as a surrogate for predictive approximation. The theory does not claim a distribution-free numerical lower bound on $\mathbf{p}_{\text{hit}}$ or a quantum advantage.

1 Introduction

Reservoir computing replaces recurrent parameter training by a fixed driven dynamical system and learns only a readout (Jaeger, 2001; Maass et al., 2002). The simplification is attractive when internal dynamics are expensive to optimize, but it moves a large part of the learning problem into representation choice: memory, reservoir width, random realization, and measurement quality all determine whether the frozen features are useful. In quantum reservoir computing (QRC), these choices are also physical resources. A useful learning theory should therefore say not only how accurately one frozen predictor is evaluated, but also why searching more frozen reservoirs or allocating more qubits can improve the predictor that is ultimately returned.

We study causal temporal regression from a single dependent realization. The target is a continuous fading-memory functional of the input history, while the learner observes only finite windows separated by a raw-time gap. Three implementation discrepancies are unavoidable: a finite window replaces the infinite past, finite quantum measurements replace exact expectation values, and a finite dependent sample replaces population risk. Two representation questions remain on top of them: what information survives the width- n preprocessor, and how likely is a finite random pool to contain a useful frozen realization?

QuaRK addresses these questions with a topology- and resource-aware QRC family. An architecture $\eta = (n, R, \lambda_+)$ fixes the qubit width, number of parallel reservoir branches, and upper memory endpoint. Gaussian preprocessing is coupled canonically across widths, without changing its fixed- n marginal law. Each branch then draws a contraction rate and one balanced random mixing circuit and freezes them. The heterogeneous rates give the parallel reservoir a multiscale memory profile. A cycle interaction graph both schedules the constant-depth entangling block and defines the measured one- and two-qubit Pauli bank. A nine-setting cover acquires that complete bank, and the resulting features feed an RMS-normalized Matérn RKHS readout with a single Ivanov radius. The kernel uses Euclidean feature distance divided by the square root of feature dimension, so increasing n or R does not change the kernel scale merely by concatenating more bounded coordinates.

Learning uses a finite frozen pool. For one architecture, S independent designs are sampled, their measured features are cached, and the design and readout are selected jointly by full-sample empirical-risk minimization. No gate axis, angle, or contraction rate is trained. Using the same sample for fitting and selection is not treated as free: the dependent-data bound is uniform over all S candidates and the complete Ivanov ball, so the adaptivity of finite-pool search is charged explicitly through $\log S$ while readout capacity is charged through Λ .

The representation analysis separates two notions of expressivity. First, the mixer analysis characterizes *finite-resource feature enrichment*: the encoding fixes the elementary trigonometric content, whereas the mixer and graph determine how that content is distributed into multiqubit observables. This does not assert that more entanglement always improves performance or that mixing enlarges the asymptotic hypothesis class. Second, for a fixed Gaussian projection, we define the best population excess risk achievable by any continuous fading-memory predictor that sees the complete projected history. This *projection-information floor* isolates what width can and cannot recover. Under the canonical Gaussian prefix coupling the floor is nonincreasing in n , and it is zero almost surely once $n \geq d$; strict improvement at every width is neither required nor claimed. We then show that finite spatial multiplexing of QuaRK coordinates followed by finite-norm Matérn readouts can approach this floor arbitrarily closely. Thus the asymptotic approximation obstruction at fixed width is exactly the information discarded by preprocessing, not an additional limitation of the frozen reservoir family.

This support-level approximation result is then connected to random search. Whenever an architecture has a strictly good support design, continuity of the causal reservoir map gives an open neighborhood of equally good designs and therefore positive random-design mass $\mathbf{p}_{\text{hit}}$. An i.i.d. pool of size S misses every such design with probability $(1 - \mathbf{p}_{\text{hit}})^S$. Combining this exact pool-coverage term with the uniform ERM oracle inequality, grouped-measurement fidelity, and contractive finite-window fidelity yields an end-to-end excess-risk bound for the operational predictor actually returned. The result gives S two transparent roles: exponentially better representation coverage and a logarithmic statistical selection price.

A finite-data Johnson–Lindenstrauss result remains useful, but for a different question. It supplies a width logarithmic in the number of raw inputs required to preserve the geometry of the particular observed sample. We keep this statement separate from the population approximation theorem rather than interpreting finite-sample geometry preservation as a prediction guarantee.

Finally, we formulate a controlled temporal benchmark that varies dependence, input dimension, memory depth, nonlinear degree, and spatiotemporal interaction structure while preserving compact inputs and a known causal target. The resource study is organized around the same quantities that appear in the theory: n , R , M , S , and λ_+ . The empirical question is not simply whether a quantum reservoir can fit a task, but which temporal processing demands justify each additional resource. State-preparation counts are kept separate from per-run circuit resources: a measured trajectory has width $O(n)$ and depth $O(w)$ under the declared reusable-hardware convention, whereas grouped acquisition requires $9RM$ branch-circuit runs per candidate window.

Contributions. The paper makes three contributions. First, it introduces a topology-aligned frozen QRC family with explicit architecture, random-design law, and grouped measurement cost, together with a graph-local operator-spreading result that explains what the nontrivial mixer and edge-observable bank

add beyond the identity support point at finite resources. Second, it gives a width-resolved approximation theorem, positive-mass and finite-pool coverage results, and a dependent-data ERM certificate that combine into an end-to-end resource-to-performance bound for the selected predictor; under geometric mixing, an explicit logarithmic-in- N gap rule keeps the coupling penalty controlled. Third, it provides a controlled evaluation framework for relating temporal-processing difficulty to (n, R, M, S, λ_+) and directly ablating mixer/observable mechanisms, while keeping finite-data JL geometry conceptually separate from population prediction.

2 Related Work

Reservoir computing and fading memory. Echo-state networks and liquid-state machines established the use of fixed driven dynamics with trained readouts (Jaeger, 2001; Maass et al., 2002). Fading memory formalizes the decreasing influence of the remote past (Boyd & Chua, 1985); universal approximation and statistical risk bounds are available for several classical reservoir families (Grigoryeva & Ortega, 2018; Gonon et al., 2020). QuaRK adopts the same separation between frozen dynamics and learned readout, but makes the physical width, reservoir multiplicity, memory range, and measurement budget explicit.

Quantum reservoirs and spatial multiplexing. QRC replaces the recurrent state by a driven quantum system and extracts classical features from quantum observables (Fujii & Nakajima, 2017; Chen & Nurdin, 2019; Chen et al., 2020). Existing constructions use resets, dissipation, multiplexing, and gate-model dynamics (Nakajima et al., 2019; Mujal et al., 2023; Molteni et al., 2023; Hu et al., 2024), while formal work studies fading-memory and universality properties of quantum reservoir filters (Martínez-Peña & Ortega, 2023; 2025; Monzani & Prati, 2025). In particular, Martínez-Peña & Ortega (2025) analyze *contracted-encoded quantum channels*: an input-encoding channel followed by a strictly contractive channel, together with conditions under which the resulting reservoir filter remains genuinely input-dependent. This design principle directly motivates the stable input-dependent channel used in QuaRK; our contribution lies in combining it with topology-aware frozen mixing, heterogeneous memory scales, grouped measurements, and resource-aware learning guarantees. We use spatial multiplexing in the standard sense of parallel uncoupled reservoirs driven by the same input. Its role here is representational: finitely many separating reservoir coordinates can coexist inside one feature map. The learned readout remains a norm-controlled Matérn RKHS rather than the polynomial class used only in the approximation argument, which also avoids importing the unfavorable qubit-dependent complexity of some polynomial-readout QRC bounds (Chmielewski et al., 2025).

Expressivity, operator spreading, and observability. Fourier analyses of parametrized quantum models show that data encoding fixes the accessible frequency set, while the remaining circuit determines which coefficient directions inside that set can actually be realized (Schuld et al., 2021). Recent QRC analysis extends this viewpoint to temporal inputs and emphasizes that increasing reservoir size or dynamical complexity alone need not increase resolvable expressive capacity when the encoding is already the bottleneck (Schütte et al., 2026). Complementary finite-sampling work formalizes resolvable expressive capacity and eigentasks, including evidence that measured correlations can improve the functions resolved above sampling noise (Hu et al., 2023). Krylov-based QRC studies instead inspect the span generated by time-evolved measured operators; Krylov observability closely tracks information-processing capacity and makes the choice of measured observables part of the expressivity question (Čindrak et al., 2025b;a).

QuaRK uses these results as motivation rather than importing their capacity claims. Our self-contained statement is narrower: product input injection cannot enlarge Pauli support, the identity mixer therefore leaves each raw observable coordinate confined to its seed qubits, whereas a generic edge-local mixer activates higher-support Pauli components whose spatial reach is controlled by graph distance. This identifies operator spreading and observable locality as concrete finite-resource mechanisms without claiming that entanglement by itself improves risk or creates frequencies absent from the encoding.

Random preprocessing and kernel readouts. Gaussian Johnson–Lindenstrauss embeddings preserve finite-set geometry at a target dimension logarithmic in the number of points (Dasgupta & Gupta, 2003; Woodruff, 2014). QuaRK uses this fact only for the raw inputs present in the observed windows; it does not

interpret finite-data projection preservation as global injectivity. The classical readout uses a normalized Matérn kernel evaluated on an RMS feature metric, equivalently on features scaled by the inverse square root of their dimension. This fixed, label-independent rescaling removes dimension-induced distance inflation while preserving the standard Matérn function class at each finite dimension. Matérn kernels are strictly positive definite and universal on compact Euclidean subsets under the standard spectral conditions (Genton, 2001; Sriperumbudur et al., 2011; Porcu et al., 2024); here those properties provide bounded RKHS complexity and the final approximation step, not a claim of quantum advantage.

Learning under temporal dependence. Independent-block coupling converts beta-mixing samples into nearby independent collections and underlies Rademacher bounds for non-i.i.d. data (Yu, 1994; Mohri & Rostamizadeh, 2008). Our sampling protocol fixes the auxiliary block length to one by constructing already-gapped windows, so the dependence term remains visible through $\beta_Z(g)$. The resulting uniform deviation is taken simultaneously over the finite frozen pool and the Ivanov ball, allowing full-sample empirical-risk selection while charging the search adaptivity explicitly through $\log S$.

Grouped quantum measurements. Large observable families can be estimated by randomized schemes such as classical shadows (Huang et al., 2020). QuaRK instead declares a small topology-local Pauli bank in advance and exploits a deterministic cover. Overlapping tomography relates such schedules to covering arrays and provides constant-size constructions for nearest-neighbor two-body Pauli marginals (Hansenne et al., 2025). Aligning the measurement bank with the interaction graph makes the feature dimension and state-preparation cost directly interpretable.

The contribution is therefore not any one of these ingredients in isolation. It is the resource-accounted combination of a frozen topology-aware QRC law, a finite-pool learning protocol, a dependent-data finite-pool ERM certificate, an explicit random-pool coverage term, and a width-resolved representation argument expressed in the same architecture/design/feature-map hierarchy.

3 Problem Setting

We first fix the causal prediction target and the dependent sample. Keeping these objects separate from the quantum model makes the later risk statements unambiguous: there is one population target, one gapped finite-window sample, and one dependence coefficient inherited from the original process.

3.1 Causal fading-memory regression

Let $Z = (Z_t)_{t \in \mathbb{Z}}$ be stationary, with $Z_t = (X_t, Y_t)$, $X_t \in \mathcal{X} \subset \mathbb{R}^d$, and $Y_t \in [-\Upsilon, \Upsilon]$. The input set $\mathcal{X}$ is compact and $\mathbb{Z}_- := \{\dots, -2, -1, 0\}$. We write $\mathbf{x}^- = (\dots, x_{-1}, x_0) \in \mathcal{X}^{\mathbb{Z}_-}$ for an input history.

The target is a measurable version $F^* : \mathcal{X}^{\mathbb{Z}_-} \rightarrow \mathbb{R}$ of the causal conditional mean,

$$F^*(X_{-\infty:0}) = \mathbb{E}[Y_0 \mid \sigma(X_t : t \leq 0)] \quad \text{a.s.}, \quad (1)$$

with Bayes risk $\mathcal{R}^* := \mathbb{E}[(F^*(X_{-\infty:0}) - Y_0)^2]$.

Let $\varpi = (\varpi_\ell)_{\ell \geq 0}$ be positive, nonincreasing, and satisfy $\varpi_\ell \rightarrow 0$. The associated fading-memory metric is

$$d_{\varpi}(\mathbf{u}^-, \mathbf{v}^-) := \sup_{\ell \geq 0} \varpi_\ell \|\mathbf{u}_{-\ell} - \mathbf{v}_{-\ell}\|_2. \quad (2)$$

A functional has the fading-memory property (FMP) when it is continuous for such a metric. More generally, for any compact $\mathcal{A} \subset \mathbb{R}^k$, write $\mathcal{F}_{\text{FMP}}(\mathcal{A})$ for the continuous fading-memory functionals on $\mathcal{A}^{\mathbb{Z}_-}$ under the analogous weighted metric. Because the weights tend to zero, this is exactly continuity for the product topology, and $\mathcal{A}^{\mathbb{Z}_-}$ is compact. We abbreviate $\mathcal{F}_{\text{FMP}} := \mathcal{F}_{\text{FMP}}(\mathcal{X})$ and equip it with the uniform norm

$$\|F\|_\infty := \sup_{\mathbf{x}^- \in \mathcal{X}^{\mathbb{Z}_-}} |F(\mathbf{x}^-)|, \quad F \in \mathcal{F}_{\text{FMP}}. \quad (3)$$

Throughout the paper, $F^* \in \mathcal{F}_{\text{FMP}}$. In particular, continuity on the compact history space implies that sufficiently long terminal windows approximate F^* uniformly.

3.2 Gapped finite windows

Only one finite realization of $\mathbf{Z}$ is observed. Following the finite-window viewpoint used in reservoir risk analysis (Gonon et al., 2020), fix a window length $w \geq 1$ and a raw-time gap $g \geq 0$, and set the stride $\Delta := w + g$. The labeled window process is

$$\mathbf{W}_j := (X_{j\Delta-w+1:j\Delta}, Y_{j\Delta}), \quad j \in \mathbb{N}. \quad (4)$$

For the dependent-data argument we take an even sample size $N = 2\mu$ and write

$$\mathcal{D}_N := (\mathbf{W}_1, \dots, \mathbf{W}_N). \quad (5)$$

The parity split therefore contains μ windows on each side. This evenness is only a proof device; empirical fitting need not otherwise privilege even sample sizes. A realized sample is denoted by $\tilde{\mathcal{D}}_N = (\tilde{\mathbf{W}}_1, \dots, \tilde{\mathbf{W}}_N)$ with $\tilde{\mathbf{W}}_j = (\mathbf{x}_{j\Delta-w+1:j\Delta}, y_{j\Delta})$.

3.3 Absolute regularity

For sigma-algebras $\mathcal{A}$ and $\mathcal{B}$, let $\beta(\mathcal{A}, \mathcal{B})$ denote their absolute-regularity coefficient. The coefficients of the original process are

$$\beta_{\mathbf{Z}}(q) := \sup_{t \in \mathbb{Z}} \beta(\sigma(\mathbf{Z}_u : u \leq t), \sigma(\mathbf{Z}_u : u \geq t + q)), \quad q \geq 0. \quad (6)$$

We call $\mathbf{Z}$ beta mixing if $\beta_{\mathbf{Z}}(q) \rightarrow 0$. Since adjacent windows are separated by the raw-time gap g , monotonicity of absolute regularity under sigma-algebra inclusion gives the only window-mixing relation used later,

$$\beta_{\mathbf{W}}(1) \leq \beta_{\mathbf{Z}}(g). \quad (7)$$

Thus the sampling protocol exposes the dependence penalty directly through g and $\beta_{\mathbf{Z}}(g)$ rather than hiding it in an effective sample size.

4 The Topology-Aware QuaRK Family

A QuaRK run separates representational choices from learning budgets. The *architecture*

$$\boldsymbol{\eta} := (n, R, \lambda_+),$$

collects the qubit width, number of parallel reservoir branches, and upper memory endpoint. The measurement repetitions M , frozen-pool size S , and Ivanov radius Λ remain external budgets: they control estimation, random search, and classical readout capacity rather than the frozen quantum representation itself. This distinction will keep the resource dependence of the later guarantees explicit without repeatedly expanding its components.

4.1 Gaussian preprocessing and interaction graph

To compare widths without changing the marginal model at any fixed n , draw an i.i.d. sequence $\mathbf{g}_1, \mathbf{g}_2, \dots \sim \mathcal{N}(0, I_d)$, independently of the temporal process, and use the canonical Gaussian prefixes

$$\boldsymbol{\Pi}_n := \frac{1}{\sqrt{n}} \begin{bmatrix} \mathbf{g}_1^\top \\ \vdots \\ \mathbf{g}_n^\top \end{bmatrix} \in \mathbb{R}^{n \times d}, \quad n \geq 3. \quad (8)$$

For every fixed width, the entries are i.i.d. $\mathcal{N}(0, 1/n)$, exactly as in a direct Gaussian draw. The coupling only gives a coherent meaning to increasing width: the narrower representation is recoverable from the wider one through

$$\boldsymbol{\Pi}_n x = \sqrt{\frac{n+1}{n}} [\boldsymbol{\Pi}_{n+1} x]_{1:n}. \quad (9)$$

A run freezes the prefix required by its chosen width. With the injective bounded angle map $\vartheta(u) := \pi \tanh u$ and $\mathbf{R}_y(\theta) := \exp(-i\theta \mathbf{Y}/2)$, encode $x \in \mathcal{X}$ by

$$\mathbf{U}_n^{\text{in}}(x) := \bigotimes_{j=1}^n \mathbf{R}_y(\vartheta((\mathbf{\Pi}_n x)_j)). \quad (10)$$

The interaction graph is the cycle $\mathcal{G}_n = (\mathcal{V}_n, \mathcal{E}_n) = C_n$, with $\mathcal{V}_n = \{1, \dots, n\}$. A minimum edge colouring writes

$$\mathcal{E}_n = \bigsqcup_{j=1}^{J_n} \mathcal{M}_{n,j}, \quad J_n = \chi'(C_n) = \begin{cases} 2, & n \text{ even}, \\ 3, & n \text{ odd}, \end{cases}$$

where each $\mathcal{M}_{n,j}$ is a matching. The same cycle determines both the entangling layers and the two-qubit observable bank.

4.2 One balanced random mixing block

Let $\boldsymbol{\sigma} = (\mathbf{X}, \mathbf{Y}, \mathbf{Z})$ and $\mathbb{S}^2 := \{\mathbf{u} \in \mathbb{R}^3 : \|\mathbf{u}\|_2 = 1\}$. For a vertex v and edge $e = \{v, v'\}$, use the local generators $\mathbf{G}_v(\mathbf{a}) := \mathbf{a} \cdot \boldsymbol{\sigma}_v$ and $\mathbf{G}_e(\mathbf{a}, \mathbf{a}') := (\mathbf{a} \cdot \boldsymbol{\sigma}_v) \otimes (\mathbf{a}' \cdot \boldsymbol{\sigma}_{v'})$.

Fix $\gamma \in (0, \pi]$. Draw the generator axes independently and uniformly on $\mathbb{S}^2$, the gate angles from an absolutely continuous law supported on $[-\gamma, \gamma]$ with positive density on its interior, and an independent uniform permutation τ of the J_n matching layers. Writing

$$\mathbf{V} := \prod_{v \in \mathcal{V}_n} \exp\left(-\frac{i}{2} \theta_v \mathbf{G}_v(\mathbf{a}_v)\right), \quad \mathbf{E}_j := \prod_{e \in \mathcal{M}_{n,j}} \exp\left(-\frac{i}{2} \theta_e \mathbf{G}_e(\mathbf{a}_e, \mathbf{a}'_e)\right),$$

the mixing unitary is

$$\mathbf{W}_n(\boldsymbol{\omega}) := \mathbf{E}_{\tau(J_n)} \cdots \mathbf{E}_{\tau(1)} \mathbf{V}, \quad \boldsymbol{\omega} = (\boldsymbol{\Gamma}, \boldsymbol{\theta}) \sim \mathbb{P}_n^\omega. \quad (11)$$

For every fixed skeleton, the zero-angle point gives $\mathbf{W}_n = \mathbf{I}$, and the unitary is real analytic in the gate angles. Its logical depth is $1 + J_n \leq 4$, so circuit depth is fixed by the family rather than treated as another model hyperparameter.

4.3 Contractive memory branches and frozen designs

Set $\lambda_0 := e^{-1}$. The architecture chooses $\lambda_+ \in (\lambda_0, 1)$, and each branch draws its contraction rate from

$$\frac{d\nu_{\lambda_+}(\lambda)}{d\lambda} = \frac{\mathbf{1}_{(\lambda_0, \lambda_+)}}{\lambda \log(\lambda_+/\lambda_0)}. \quad (12)$$

Thus equal multiplicative rate intervals receive equal probability mass and λ_+ alone sets the longest available memory scale. The branchwise randomization has a representational purpose beyond enforcing contraction: a state difference propagated through branch r for ℓ additional steps is attenuated by λ_r^ℓ , so the natural e-folding horizon is $(-\log \lambda_r)^{-1}$. Drawing distinct rates across the R parallel branches therefore creates a multiscale memory bank, from rapidly forgetting to more persistent dynamics. This heterogeneity increases the chance that a frozen multiplex contains temporal responses compatible with the fading-memory profile of the target functional; it is a design motivation rather than a finite-pool performance guarantee.

Let $\rho_+ := |+\rangle\langle +|^{\otimes n}$, with $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$, and define the replacement channel $\mathcal{P}_+(\mathbf{Q}) := \text{Tr}[\mathbf{Q}]\rho_+$. For $r = 1, \dots, R$, draw independently $(\lambda_r, \boldsymbol{\omega}_r) \sim \nu_{\lambda_+} \otimes \mathbb{P}_n^\omega$ and set

$$\mathcal{U}_{r,x}(\mathbf{Q}) := \mathbf{W}_n(\boldsymbol{\omega}_r) \mathbf{U}_n^{\text{in}}(x) \mathbf{Q} \mathbf{U}_n^{\text{in}}(x)^\dagger \mathbf{W}_n(\boldsymbol{\omega}_r)^\dagger.$$

The driven reservoir channel is

$$\mathcal{T}_{r,x}(\mathbf{Q}) := \lambda_r \mathcal{U}_{r,x}(\mathbf{Q}) + (1 - \lambda_r) \mathcal{P}_+(\mathbf{Q}), \quad (13)$$

and is strictly contractive because $\lambda_r < 1$. Equivalently, this is an input-encoding channel followed by the fixed contraction $\mathbf{Q} \mapsto \lambda_r \mathbf{W}_n(\omega_r) \mathbf{Q} \mathbf{W}_n(\omega_r)^\dagger + (1 - \lambda_r) \mathcal{P}_+(\mathbf{Q})$. This follows the contracted-encoded QRC design principle studied by [Martínez-Peña & Ortega \(2025\)](#), where strict contraction supplies the echo-state and fading-memory mechanism while input dependence is carried by the encoding; their work further analyzes when such filters distinguish input sequences. QuaRK uses this mechanism branchwise and adds topology-aware random mixing and the multiscale spatial multiplex described above.

For the architecture η , the frozen-design law is

$$\mathbb{P}_\eta := (\nu_{\lambda_+} \otimes \mathbb{P}_n^\omega)^{\otimes R}, \quad \mathbf{D} \sim \mathbb{P}_\eta. \quad (14)$$

Here $\mathbf{D} = ((\lambda_r, \omega_r))_{r=1}^R$ records the branch-specific parameters; the already-drawn $\mathbf{\Pi}_n$ is shared across all branches. Once sampled, no quantum parameter is trained.

4.4 Topology-local observables and grouped measurements

The complete observable bank contains all one-qubit Paulis and all two-qubit Pauli products on graph edges:

$$\begin{aligned} \mathcal{O}_V(n) &:= \{\mathbf{X}_v, \mathbf{Y}_v, \mathbf{Z}_v : v \in \mathcal{V}_n\}, \\ \mathcal{O}_E(n) &:= \{\mathbf{P}_u \mathbf{P}'_v : \{u, v\} \in \mathcal{E}_n, \mathbf{P}, \mathbf{P}' \in \{\mathbf{X}, \mathbf{Y}, \mathbf{Z}\}\}, \\ \mathcal{O}_n &:= \mathcal{O}_V(n) \cup \mathcal{O}_E(n), \quad q_n := |\mathcal{O}_n| = 12n. \end{aligned} \quad (15)$$

Let $\mathcal{S}_n = \{s_k : k = 1, \dots, K_n\}$ be a Pauli-setting cover and let $a_n(O)$ count the settings compatible with O . For the cycle, the nine-setting construction of Section A.4 gives $K_n = 9$, with $a_n(O) = 3$ for $O \in \mathcal{O}_V(n)$ and $a_n(O) = 1$ for $O \in \mathcal{O}_E(n)$. We use a reusable- n -qubit hardware convention for the acquisition count: repeating every setting M times for each branch estimates the entire bank and costs $RK_nM = 9RM$ branch-circuit executions/state preparations per input window.

This count measures independent circuit executions/state preparations, not the depth or width of one execution. Each execution propagates one n -qubit branch through the w inputs in the window. Under the declared parallel gate schedule, the per-step input injection, contraction/reset operation, and balanced mixer have constant logical depth, so the full trajectory depth is $O(w)$ and the quantum width is $O(n)$ under this branch-by-branch convention. If all R branches are instead physically instantiated at once, the width is $O(Rn)$ and up to the factor R can be traded from wall-clock circuit executions into parallel hardware. The paper reports the reusable-hardware run count so that R has one unambiguous acquisition interpretation.

4.5 What the mixer contributes: operator spreading and direct interactions

The universality argument later in the paper deliberately uses the identity support point of the random circuit law. That certificate is useful for showing that the family has no asymptotic approximation floor, but by itself it does not explain why the nontrivial mixer in Equation 11 should be retained. The finite-resource role of the mixer is different: it spreads local observables across the interaction graph before measurement and thereby exposes multiqubit functions of the encoded input directly in the frozen feature vector.

For a Pauli string P , write $\text{supp}(P)$ for its nonidentity sites and $\text{wt}(P) := |\text{supp}(P)|$. If a Hermitian operator $A = \sum_P a_P P$ is expanded in the n -qubit Pauli basis, define its Pauli interaction order by

$$\text{pord}(A) := \max\{\text{wt}(P) : a_P \neq 0\}. \quad (16)$$

For $A \subseteq \mathcal{V}_n$, let $\mathcal{N}_{C_n}^\ell(A) := \{v : \text{dist}_{C_n}(v, A) \leq \ell\}$.

Proposition 1 (Mixer-free locality). *Set $\mathbf{W}_n = \mathbf{I}$. If a measured Pauli observable O is supported on $A \subseteq \mathcal{V}_n$, then its finite-window and causal reservoir coordinates depend on the input history only through the projected scalar streams*

$$((\mathbf{\Pi}_n x_t)_j : t \leq 0, j \in A).$$

Consequently, vertex observables expose only one projected stream directly, and the full bank $\mathcal{O}_n$ exposes at most two projected streams directly in any single raw feature coordinate.

Theorem 2 (Graph-local operator spreading). *Let O be a Pauli string supported on A . For every realization of the balanced circuit, every Pauli component of $\mathbf{W}_n(\omega)^\dagger O \mathbf{W}_n(\omega)$ is supported inside $\mathcal{N}_{C_n}^{J_n}(A)$. After s repetitions of the same mixing block, the support is contained in $\mathcal{N}_{C_n}^{sJ_n}(A)$.*

Moreover, the declared continuous random-circuit law activates nonlocal components generically. For every oriented cycle edge $u \rightarrow v$, the coefficient of $\mathbf{Z}_u \mathbf{Z}_v$ in $\mathbf{W}_n^\dagger \mathbf{X}_u \mathbf{W}_n$ is nonzero with probability one. For every length-two path $u - v - w$ with distinct vertices, the coefficient of $\mathbf{X}_u \mathbf{Z}_v \mathbf{Z}_w$ in $\mathbf{W}_n^\dagger (\mathbf{X}_u \mathbf{X}_v) \mathbf{W}_n$ is nonzero with probability one.

Corollary 3 (Direct interaction enrichment). *Let $\alpha_j(x) := \vartheta((\Pi_n x)_j)$. For a one-step branch started from ρ_+ , a generic QuaRK mixer gives the $\mathbf{X}_u$ feature a nonzero term proportional to $\sin \alpha_u(x) \sin \alpha_v(x)$ for every neighbor v of u . Likewise, an edge feature $\mathbf{X}_u \mathbf{X}_v$ has, for either neighboring site w outside that edge, a nonzero term proportional to $\cos \alpha_u(x) \sin \alpha_v(x) \sin \alpha_w(x)$. Thus the edge bank seeds genuine three-stream interactions after one mixing block, whereas the identity mixer is restricted to direct interaction order at most two.*

Interpretation. The theorem does not claim that entanglement automatically lowers prediction risk, nor that mixing creates frequencies absent from the encoding. Rather, the local encoding fixes the elementary one-coordinate trigonometric factors, while the mixer changes which multivariate coefficient directions are exposed to the readout. This distinction is consistent with Fourier-based QML and QRC expressivity analyses, where the encoding determines the accessible frequency structure while the reservoir dynamics and readouts determine how much of that structure is realized (Schuld et al., 2021; Schütte et al., 2026). It is also aligned with recent Krylov-observability work, which relates the span of time-evolved observables and the choice of measured operators to information processing capacity (Čindrak et al., 2025b;a).

The cycle now has a precise resource role. It is connected, so it imposes no permanent decomposition into isolated components, while its bounded degree permits every entangling block to be scheduled in $J_n \leq 3$ matching layers. The graph controls the light cone in which interactions can be generated; the edge observables provide two-site seeds and therefore expose one higher direct interaction order than vertex observables after a generic block. These are finite-resource representation statements. They do not contradict the later identity-support universality proof, whose universal nonlinear readout can compose simpler coordinates given sufficiently many resources.

4.6 Exact and measured feature maps

For a finite window $\mathbf{x} = x_{-w+1:0}$, the state of branch r is

$$\rho_r^D(\mathbf{x}) := \mathcal{T}_{r, x_0} \circ \cdots \circ \mathcal{T}_{r, x_{-w+1}}(\rho_+).$$

Strict contraction gives the corresponding causal state $\rho_{r, \infty}^D(\mathbf{x}^-)$ by moving the initialization to the remote past. The branch feature block is $\phi_r^D(\mathbf{x}) := (\text{Tr}[O \rho_r^D(\mathbf{x})])_{O \in \mathcal{O}_n}$. With $m_\eta := Rq_n$, the exact finite-window feature map is

$$\Phi_D(\mathbf{x}) := (\phi_1^D(\mathbf{x}), \dots, \phi_R^D(\mathbf{x})) \in [-1, 1]^{m_\eta}, \quad (17)$$

and $\Phi_{D, \infty}$ is defined analogously from the causal states. We write $\hat{\Phi}_D(\mathbf{x}; \chi)$ for the estimator obtained from the grouped measurement record χ . Exact and measured features remain distinct throughout the analysis.

4.7 Matérn readout and risks

Feature dimension changes with the architecture, so an unnormalized Euclidean kernel distance would mechanically grow when more bounded coordinates are concatenated. QuaRK removes that nuisance at the model-definition level. For $z, z' \in \mathbb{R}^p$, define the root-mean-square feature distance

$$d_p^{\text{rms}}(z, z') := \frac{\|z - z'\|_2}{\sqrt{p}}. \quad (18)$$

Let κ_p be the normalized Matérn kernel with smoothness $\nu > 1$ and length scale $\xi > 0$ evaluated at $d_p^{\text{rms}}(z, z')$. To remove any implementation ambiguity, throughout the paper “Matérn(ν, ξ)” means

$$\kappa_p(z, z') := \frac{2^{1-\nu}}{\Gamma(\nu)} \left(\frac{\sqrt{2\nu} d_p^{\text{rms}}(z, z')}{\xi} \right)^\nu K_\nu \left(\frac{\sqrt{2\nu} d_p^{\text{rms}}(z, z')}{\xi} \right), \quad \kappa_p(z, z) := 1, \quad (19)$$

where K_ν is the modified Bessel function of the second kind, and let $\mathcal{H}_p$ be its RKHS. Equivalently, this is the ordinary fixed- (ν, ξ) Matérn kernel applied to $z/\sqrt{p}$. The $1/\sqrt{p}$ factor is fixed once by the method, is label independent, and is never retuned across architectures. For every finite p the rescaling is invertible, so it does not remove information or alter Matérn universality on compact feature images.

The Ivanov ball is $\mathcal{H}_p^\Lambda := \{h \in \mathcal{H}_p : \|h\|_{\mathcal{H}_p} \leq \Lambda\}$. Kernel normalization gives $|h(z)| \leq \Lambda$, while the section map is Lipschitz in the RMS metric with the dimension-free constant

$$L_\kappa := \frac{1}{\xi} \sqrt{\frac{\nu}{\nu-1}}.$$

Equivalently, its Lipschitz constant with respect to the raw Euclidean feature vector is $L_\kappa/\sqrt{p}$. This normalization prevents feature count alone from being interpreted as either architectural improvement or degradation; the empirical protocol keeps one fixed Matérn specification and checks the resulting similarity geometry directly. The radius Λ is fixed before fitting; Section 6.3 provides a direct calibration from a requested generalization tolerance.

For a frozen design D , define the causal, finite-window, and operational risks together by

$$\begin{aligned} \mathcal{R}_{D,\infty}(h) &:= \mathbb{E}[(h(\Phi_{D,\infty}(X_{-\infty:0})) - Y_0)^2], \\ \mathcal{R}_{D,w}(h) &:= \mathbb{E}[(h(\Phi_D(X_{-w+1:0})) - Y_0)^2], \\ \mathcal{R}_{D,w}^{\text{op}}(h) &:= \mathbb{E}[(h(\hat{\Phi}_D(X_{-w+1:0})) - Y_0)^2], \end{aligned}$$

where the last expectation also averages over fresh measurement outcomes. For nonempty $\mathcal{I} \subseteq [N]$, the measured empirical risk is

$$\hat{\mathcal{R}}_{D,w}^{\mathcal{I}}(h) := \frac{1}{|\mathcal{I}|} \sum_{j \in \mathcal{I}} (h(\hat{\Phi}_D(X_{j\Delta-w+1:j\Delta})) - Y_{j\Delta})^2, \quad \hat{\mathcal{R}}_{D,w} := \hat{\mathcal{R}}_{D,w}^{[N]}.$$

Finally, make the readout-radius dependence of the causal approximation error explicit:

$$\varepsilon_{\text{app}}(D; \Lambda) := \inf_{h \in \mathcal{H}_{m_\eta}^\Lambda} \{\mathcal{R}_{D,\infty}(h) - \mathcal{R}^*\}. \quad (20)$$

This quantity links the width-resolved representation analysis in Section 6.4 to the finite-pool performance theorem.

5 Finite Frozen-Pool ERM

Fix an architecture η together with measurement budget M , pool size S , and readout radius Λ . Each candidate runs the same declared architecture but with an independently frozen realization. Selection is over this finite pool only: no gate axis, angle, or contraction rate is optimized.

5.1 Frozen pool and cached measurements

Draw

$$D_1, \dots, D_S \stackrel{\text{i.i.d.}}{\sim} \mathbb{P}_\eta,$$

independently of the labeled sample and freeze the pool. For every observed window and every candidate s , generate $\hat{\Phi}_{D_s}$ once and cache it. With the cycle cover, the complete measured training pool costs $9NMSR$ independent state preparations/branch-circuit runs under the reusable- n -qubit convention; all subsequent readout fits and design selection are classical. As clarified in Section 4.4, this acquisition count is separate from the $O(w)$ logical depth and $O(n)$ width of one trajectory.

5.2 Joint empirical-risk selection

QuaRK selects the frozen design and its readout by one finite-pool empirical risk minimization,

$$(\hat{s}, h^*) \in \arg \min_{\substack{1 \leq s \leq S \\ h \in \mathcal{H}_{m_\eta}^\Lambda}} \hat{\mathcal{R}}_{D_s, w}(h), \quad D^* := D_{\hat{s}}, \quad (21)$$

with a deterministic candidate tie rule and the minimum-RKHS-norm minimizer within a tied candidate. Equivalently, one fits an Ivanov-regularized kernel least-squares readout for each frozen design and keeps the smallest measured empirical risk. The representer theorem reduces every fit to a finite convex problem (Page & Grünewälder, 2019; Schölkopf et al., 2001).

Using the same sample for fitting and selection is deliberate rather than an assumption that overfitting cannot occur. The uniform deviation result in Section 6.3 holds simultaneously over all S designs and the entire Ivanov ball. It therefore controls this data-dependent ERM directly: searching more frozen designs costs only a $\log S$ term, while readout adaptivity is controlled by Λ . This makes the statistical price of frozen-pool search explicit without introducing a second validation procedure.

The returned predictor is $h^* \circ \hat{\Phi}_{D^*}$ operationally, with $h^* \circ \Phi_{D^*, \infty}$ as its causal exact-feature counterpart. The role of S is quantified in Section 6.5: it increases the chance that the frozen pool contains a near-best representation, while the same increase is charged statistically through the uniform $\log S$ term.

6 Learning and Representation Guarantees

The analysis follows the actual learning pipeline. We first control the three errors that arise even for a fixed frozen representation: finite-window initialization, finite-shot measurement, and dependent finite data. We then resolve the representational role of width n by isolating the information lost by the shared Gaussian projection. This width-resolved approximation result gives positive probability to neighborhoods of good frozen designs, which in turn makes the role of the pool size S explicit. The final theorem combines these ingredients for the full-sample ERM predictor returned by Section 5.

6.1 Memory fidelity

Lemma 4 (Exact trace-norm contraction). *Every branch with rate λ satisfies*

$$\|\mathcal{T}_x(\rho) - \mathcal{T}_x(\rho')\|_1 = \lambda \|\rho - \rho'\|_1 \leq 2\lambda. \quad (22)$$

Hence every input history determines a unique causal state, and two trajectories driven by the same final w inputs differ by at most $2\lambda^w$ in trace norm.

Proposition 5 (Finite-window risk error). *For every frozen design $D \sim \mathbb{P}_\eta$ and every $h \in \mathcal{H}_{m_\eta}^\Lambda$,*

$$\begin{aligned} |\mathcal{R}_{D, \infty}(h) - \mathcal{R}_{D, w}(h)| &\leq 4\Lambda(\Lambda + \Upsilon)L_\kappa \left(\frac{1}{R} \sum_{r=1}^R \lambda_r^{2w} \right)^{1/2} \\ &\leq \underbrace{4\Lambda(\Lambda + \Upsilon)L_\kappa \lambda_+^w}_{\varepsilon_{\text{win}}(\eta)}. \end{aligned} \quad (23)$$

Corollary 6 (Memory-endpoint calibration). *A requested tolerance $\epsilon_W > 0$ is guaranteed whenever*

$$\lambda_0 < \lambda_+ \leq \min \left\{ 1, \left[\frac{\epsilon_W}{4\Lambda(\Lambda + \Upsilon)L_\kappa} \right]^{1/w} \right\}. \quad (24)$$

If the right-hand upper bound is at most λ_0 , the requested fidelity is incompatible with the fixed one-step lower memory endpoint.

Interpretation. Larger λ_+ retains longer memory but slows convergence from the finite-window implementation to its causal limit. The calibration therefore turns a desired memory-fidelity tolerance into an explicit admissible endpoint. The RMS Matérn geometry averages coordinatewise feature displacement, so merely concatenating more bounded branches or observables does not worsen this risk-sensitivity bound. It does not claim that longer memory is always more predictive; that question belongs to the representation term introduced below.

6.2 Measurement fidelity

Proposition 7 (Grouped-measurement risk error). *For every frozen design and every $h \in \mathcal{H}_{m_\eta}^\Lambda$,*

$$\begin{aligned} \left| \mathcal{R}_{D,w}^{\text{op}}(h) - \mathcal{R}_{D,w}(h) \right| &\leq 2\Lambda(\Lambda + \Upsilon)L_\kappa \left[\frac{1}{Rq_n} \mathbb{E} \sum_{r=1}^R \sum_{O \in \mathcal{O}_n} \frac{1 - \phi_{r,O}(X_{-w+1:0})^2}{a_n(O)M} \right]^{1/2} \\ &\leq 2\Lambda(\Lambda + \Upsilon)L_\kappa \sqrt{\frac{1}{q_n M} \sum_{O \in \mathcal{O}_n} \frac{1}{a_n(O)}} =: \varepsilon_{\text{shot}}(\boldsymbol{\eta}, M). \end{aligned} \quad (25)$$

For the cycle cover, $\varepsilon_{\text{shot}}(\boldsymbol{\eta}, M) = 2\Lambda(\Lambda + \Upsilon)L_\kappa \sqrt{5/(6M)}$.

Corollary 8 (Measurement-budget calibration). *For the cycle cover, $\varepsilon_{\text{shot}}(\boldsymbol{\eta}, M) \leq \epsilon_M$ follows from*

$$M \geq \left\lceil \frac{10\Lambda^2(\Lambda + \Upsilon)^2 L_\kappa^2}{3\epsilon_M^2} \right\rceil. \quad (26)$$

Interpretation. The sufficient repetitions per Pauli setting scale as $M = O(\epsilon_M^{-2})$ with no artificial Rn factor: RMS kernel geometry prevents a larger feature vector from amplifying coordinatewise estimation noise solely through dimension. This does *not* make a larger quantum representation free. The complete bank still requires $9RM$ independent branch-state preparations/circuit runs per input window under the reusable n -qubit convention, and each run uses an n -qubit trajectory of depth $O(w)$ as specified in Section 4.4.

6.3 Dependent-data generalization and ERM capacity

Condition on the shared preprocessor $\boldsymbol{\Pi}_n$ and the S frozen designs. Let $\mathbf{M}_j$ collect all candidate measurement records for window j and set $\mathbf{U}_j := (\mathbf{W}_j, \mathbf{M}_j)$. Independent measurement kernels preserve the one-step absolute-regularity coefficient, so

$$\beta_{\mathbf{U}}(1) = \beta_{\mathbf{W}}(1) \leq \beta_{\mathbf{Z}}(g). \quad (27)$$

For confidence $\delta \in (0, 1)$, write $\delta_g := \delta - 4(\mu - 1)\beta_{\mathbf{Z}}(g)$ and assume $\delta_g > 0$. Define

$$\varepsilon_{\text{gen}}(\Lambda, S, \delta) := \frac{4\Lambda(\Lambda + \Upsilon)}{\sqrt{\mu}} + 3(\Lambda + \Upsilon)^2 \sqrt{\frac{\log(4S/\delta_g)}{2\mu}}. \quad (28)$$

Proposition 9 (Uniform measured-to-operational deviation). *Conditional on any realized $\boldsymbol{\Pi}_n$ and frozen pool, with probability at least $1 - \delta$ over the gapped sample and all training measurement outcomes, simultaneously for every $s \leq S$ and every $h \in \mathcal{H}_{m_\eta}^\Lambda$,*

$$\left| \widehat{\mathcal{R}}_{D_s,w}(h) - \mathcal{R}_{D_s,w}^{\text{op}}(h) \right| \leq \varepsilon_{\text{gen}}(\Lambda, S, \delta). \quad (29)$$

Corollary 10 (Geometric-mixing gap rule). *Assume that the observed process has a known geometric absolute-regularity envelope*

$$\beta_{\mathbf{Z}}(q) \leq C_\beta e^{-c_\beta q}, \quad q \geq 0, \quad (30)$$

for constants $C_\beta \geq 1$ and $c_\beta > 0$. For $\mu \geq 2$, the choice

$$g \geq \left\lceil \frac{1}{c_\beta} \log \left(\frac{8C_\beta(\mu - 1)}{\delta} \right) \right\rceil \quad (31)$$

ensures $\delta_g \geq \delta/2$. Hence proposition 9 holds with the explicit upper bound

$$\varepsilon_{\text{gen}}(\Lambda, S, \delta) \leq \frac{4\Lambda(\Lambda + \Upsilon)}{\sqrt{\mu}} + 3(\Lambda + \Upsilon)^2 \sqrt{\frac{\log(8S/\delta)}{2\mu}}. \quad (32)$$

At fixed confidence, a raw-time gap of order $\log N$ is therefore sufficient to keep the unit-block coupling penalty controlled under geometric mixing.

For a requested uniform-deviation tolerance ϵ_G , abbreviate

$$a := \frac{4}{\sqrt{\mu}}, \quad b := 3\sqrt{\frac{\log(4S/\delta_g)}{2\mu}},$$

so that $\varepsilon_{\text{gen}}(\Lambda, S, \delta) = (a+b)\Lambda^2 + (a+2b)\Upsilon\Lambda + b\Upsilon^2$.

Corollary 11 (Readout-capacity calibration). *If $\epsilon_G > b\Upsilon^2$, the unique positive radius satisfying $\varepsilon_{\text{gen}}(\Lambda, S, \delta) = \epsilon_G$ is*

$$\Lambda = \frac{-(a+2b)\Upsilon + \sqrt{(a+2b)^2\Upsilon^2 + 4(a+b)(\epsilon_G - b\Upsilon^2)}}{2(a+b)}. \quad (33)$$

If $\epsilon_G \leq b\Upsilon^2$, no positive Ivanov radius attains the requested uniform tolerance with the declared (N, g, S, δ) .

Why full-sample ERM remains controlled. The deviation is simultaneous over the entire finite pool and the entire Ivanov ball. Therefore the data-dependent minimizer in Equation 21 is already covered by the same event; no independent validation fold is required for the theorem. Searching more frozen designs is charged through $\log S$, while fitting a richer readout is charged through Λ .

6.4 What qubit width can improve: the projection-information floor

A finite-data Johnson–Lindenstrauss statement cannot by itself imply lower prediction risk. The correct target-sensitive role of n is the information retained by Π_n . For a fixed realization of the coupled Gaussian prefixes in Equation 8, let $\Pi_n^{\mathbb{Z}^-}$ act coordinatewise on histories and define

$$\mathcal{F}_n^\Pi := \{G \circ \Pi_n^{\mathbb{Z}^-} : G \in \mathcal{F}_{\text{FMP}}(\Pi_n \mathcal{X})\}, \quad \mathfrak{J}_n^\Pi := \mathbb{E} \left[(F^*(X_{-\infty:0}) - \mathbb{E}[F^*(X_{-\infty:0}) \mid \sigma(\Pi_n X_t : t \leq 0)])^2 \right]. \quad (34)$$

This is the information-theoretic excess risk caused by observing only the projected history. Because $(\Pi_n \mathcal{X})^{\mathbb{Z}^-}$ is compact metrizable and continuous functions are dense in L^2 of every Borel probability on it, our FMP convention also gives the equivalent variational form

$$\mathfrak{J}_n^\Pi = \inf_{F \in \mathcal{F}_n^\Pi} \mathbb{E}[(F(X_{-\infty:0}) - F^*(X_{-\infty:0}))^2]. \quad (35)$$

Thus the floor isolates projection information loss before any restriction to a particular reservoir realization.

Why the approximation proof may use $W_n = \mathbf{I}$. The separation argument below uses the identity support point because it gives an especially transparent universality certificate. This should not be read as a claim that mixing is redundant at finite resources. The two questions are logically different: Section 4.5 proves that a generic graph-local mixer exposes two- and three-stream interactions directly in raw observable coordinates that the identity dynamics cannot expose at the same observable locality, whereas the present section asks only whether the support-closure family with an unrestricted universal readout has an intrinsic approximation floor. A universal readout can eventually compose local coordinates even when the mixer is set to the identity; the mixer theorem addresses how much interaction structure is already present before that classical composition.

Lemma 12 (Fading memory and projected-history separation). *Fix $n \geq 3$ and $\lambda_+ \in (\lambda_0, 1)$. Every causal coordinate of an admissible single-reservoir design is a continuous fading-memory functional of the projected history. Moreover, for every two distinct projected histories $\mathbf{z}^- \neq \mathbf{z}'^-$ in $(\Pi_n \mathcal{X})^{\mathbb{Z}^-}$, there exists a single-reservoir design in the support of the declared law whose causal feature vector separates them.*

Theorem 13 (Width-resolved QuaRK approximation). *Fix a realized preprocessor $\mathbf{\Pi}_n$ and $\lambda_+ \in (\lambda_0, 1)$. With $\boldsymbol{\eta}_R := (n, R, \lambda_+)$,*

$$\lim_{\Lambda \rightarrow \infty} \inf_{R \geq 1} \inf_{\mathbf{D} \in \text{supp}(\mathbb{P}_{\boldsymbol{\eta}_R})} \varepsilon_{\text{app}}(\mathbf{D}; \Lambda) = \mathfrak{J}_n^\Pi. \quad (36)$$

Equivalently, finite spatial multiplexing and finite-norm Matérn readouts have no asymptotic approximation floor beyond the information discarded by $\mathbf{\Pi}_n$ itself.

Corollary 14 (Monotone width and the injective regime). *Under the canonical Gaussian coupling of Equation 8,*

$$\mathfrak{J}_{n+1}^\Pi \leq \mathfrak{J}_n^\Pi, \quad n \geq 3. \quad (37)$$

If $n \geq d$, then $\mathbf{\Pi}_n$ has full column rank almost surely and

$$\mathfrak{J}_n^\Pi = 0 \quad \text{almost surely.} \quad (38)$$

No strict improvement is claimed at every increment of n : a target that already factors through a narrower projection can have the same floor at several widths.

Interpretation. This is the predictive role of width that the finite-data JL lemma alone cannot supply. Wider coupled projections never remove information available at a narrower width, while $n \geq d$ eliminates the projection bottleneck for a full-column-rank Gaussian map. The theorem does not say that finite-sample risk must decrease monotonically with n : the newly retained information may be irrelevant to the target, and larger width still consumes more physical qubits even though the RMS readout prevents coordinate count alone from inflating the fidelity bounds.

6.5 Good-design mass and finite-pool coverage

The width-resolved theorem is an existence statement over the support. To connect it to random search, define the conditional probability mass of designs whose causal approximation error lies within $\varepsilon_{\text{rep}} > 0$ of the projection floor,

$$\mathfrak{p}_{\boldsymbol{\eta}}(\varepsilon_{\text{rep}}; \Lambda \mid \mathbf{\Pi}_n) := \mathbb{P}_{\boldsymbol{\eta}}(\varepsilon_{\text{app}}(\mathbf{D}; \Lambda) \leq \mathfrak{J}_n^\Pi + \varepsilon_{\text{rep}}). \quad (39)$$

The notation emphasizes that the design law is sampled after the shared preprocessor has been frozen; the numerical value is target- and projection-dependent.

Proposition 15 (Support point implies positive good-design mass). *Fix $\boldsymbol{\eta}$, Λ , and $\varepsilon_{\text{rep}} > 0$. If some $\mathbf{D}_0 \in \text{supp}(\mathbb{P}_{\boldsymbol{\eta}})$ satisfies*

$$\varepsilon_{\text{app}}(\mathbf{D}_0; \Lambda) < \mathfrak{J}_n^\Pi + \varepsilon_{\text{rep}}, \quad (40)$$

then $\mathfrak{p}_{\boldsymbol{\eta}}(\varepsilon_{\text{rep}}; \Lambda \mid \mathbf{\Pi}_n) > 0$. Consequently, by theorem 13, for every $\varepsilon_{\text{rep}} > 0$ there exist finite R and finite Λ for which the good-design mass is positive.

Proposition 16 (Finite-pool coverage). *Let $\mathfrak{p}_{\text{hit}} := \mathfrak{p}_{\boldsymbol{\eta}}(\varepsilon_{\text{rep}}; \Lambda \mid \mathbf{\Pi}_n)$. Conditional on $\mathbf{\Pi}_n$, an i.i.d. pool of size S satisfies*

$$\mathbb{P} \left[\min_{1 \leq s \leq S} \varepsilon_{\text{app}}(\mathbf{D}_s; \Lambda) > \mathfrak{J}_n^\Pi + \varepsilon_{\text{rep}} \mid \mathbf{\Pi}_n \right] = (1 - \mathfrak{p}_{\text{hit}})^S. \quad (41)$$

Interpretation. The pool size now has a direct theoretical role: it suppresses the probability of missing an ε_{rep} -good frozen representation exponentially fast. The paper proves positivity of $\mathfrak{p}_{\text{hit}}$ when the architecture has a strictly good support point, but it does not turn Equation 41 into an a-priori numerical calibration of S . Such a calibration would require a useful quantitative lower bound on $\mathfrak{p}_{\text{hit}}$, which is target-, projection-, and architecture-dependent and is not available under the present assumptions. Campaign III therefore measures the empirical leverage of S rather than treating a formal inversion of the unknown hit mass as an operational prescription.

6.6 ERM oracle inequality and end-to-end performance

Theorem 17 (Finite-pool ERM oracle inequality). *If $\delta_g > 0$, then conditional on any realized $\mathbf{\Pi}_n$ and frozen pool, with probability at least $1 - \delta$ over the gapped sample and training measurements,*

$$\mathcal{R}_{\mathcal{D}^*,w}^{\text{op}}(h^*) \leq \min_{1 \leq s \leq S} \inf_{h \in \mathcal{H}_{m_\eta}^\Lambda} \mathcal{R}_{\mathcal{D}^*,w}^{\text{op}}(h) + 2\varepsilon_{\text{gen}}(\Lambda, S, \delta). \quad (42)$$

The inequality is precisely the anti-overfitting role of the uniform bound: the design and readout are selected on the full sample, but the price of doing so is explicit and finite.

Theorem 18 (Resource-to-performance certificate). *Fix $\boldsymbol{\eta} = (n, R, \lambda_+)$, M, S, Λ , and $\varepsilon_{\text{rep}} > 0$, and set $\mathbf{p}_{\text{hit}} := \mathbf{p}_{\boldsymbol{\eta}}(\varepsilon_{\text{rep}}; \Lambda \mid \mathbf{\Pi}_n)$. If $\delta_g > 0$, then conditional on the realized shared projection, jointly over the frozen pool, dependent sample, and training measurements, with probability at least*

$$1 - \delta - (1 - \mathbf{p}_{\text{hit}})^S, \quad (43)$$

we have

$$\boxed{\mathcal{R}_{\mathcal{D}^*,w}^{\text{op}}(h^*) - \mathcal{R}^* \leq \mathfrak{J}_n^\Pi + \varepsilon_{\text{rep}} + 2\varepsilon_{\text{gen}}(\Lambda, S, \delta) + \varepsilon_{\text{shot}}(\boldsymbol{\eta}, M) + \varepsilon_{\text{win}}(\boldsymbol{\eta})}. \quad (44)$$

On the same event, the causal exact-feature counterpart obeys

$$\mathcal{R}_{\mathcal{D}^*,\infty}(h^*) - \mathcal{R}^* \leq \mathfrak{J}_n^\Pi + \varepsilon_{\text{rep}} + 2\varepsilon_{\text{gen}}(\Lambda, S, \delta) + 2\varepsilon_{\text{shot}}(\boldsymbol{\eta}, M) + 2\varepsilon_{\text{win}}(\boldsymbol{\eta}). \quad (45)$$

How to read the theorem. The terms have distinct roles. Width n controls the irreducible information floor $\mathfrak{J}_n^\Pi$; finite R and Λ determine whether the design law places positive mass within ε_{rep} of that floor; S exponentially reduces the probability of missing such a design but simultaneously appears through $\log S$ in the statistical term; M controls measurement fidelity; and (w, λ_+) control finite-window fidelity. Unlike a conjunction of unrelated certificates, every term in Equation 44 participates in the risk of the predictor that the algorithm actually returns.

Corollary 19 (Operational-budget calibration at fixed pool size). *Fix tolerances $\varepsilon_{\text{rep}}, \epsilon_G, \epsilon_M, \epsilon_W > 0$ and a declared pool size S . For fixed $\boldsymbol{\eta}, \Lambda$, and realized $\mathbf{\Pi}_n$, suppose $\mathbf{p}_{\text{hit}} > 0$ and $\delta_g > 0$. If*

$$\begin{aligned} \varepsilon_{\text{gen}}(\Lambda, S, \delta) &\leq \epsilon_G, \\ M &\geq \left\lceil \frac{10\Lambda^2(\Lambda + \Upsilon)^2 L_\kappa^2}{3\epsilon_M^2} \right\rceil, \\ \lambda_+ &\leq \min \left\{ 1, \left[\frac{\epsilon_W}{4\Lambda(\Lambda + \Upsilon)L_\kappa} \right]^{1/w} \right\}, \end{aligned} \quad (46)$$

then, conditional on $\mathbf{\Pi}_n$, with probability at least $1 - \delta - (1 - \mathbf{p}_{\text{hit}})^S$,

$$\mathcal{R}_{\mathcal{D}^*,w}^{\text{op}}(h^*) - \mathcal{R}^* \leq \mathfrak{J}_n^\Pi + \varepsilon_{\text{rep}} + 2\epsilon_G + \epsilon_M + \epsilon_W. \quad (47)$$

For fixed (N, g, S, δ) , corollary 11 supplies the corresponding admissible Ivanov radius for a requested ϵ_G . This corollary calibrates the quantities whose bounds are explicit; it deliberately does not prescribe S from the unknown target-specific hit mass.

6.7 Complementary finite-data preprocessing geometry

The preceding theorem gives the predictive meaning of width. A separate finite-data question is how wide a Gaussian preprocessor must be to preserve the geometry of the raw inputs that actually appear in the observed windows. Let $\mathfrak{P}_N$ be their distinct values. Since $|\mathfrak{P}_N| \leq Nw$, the width can be chosen without inspecting their realized geometry.

Proposition 20 (Finite-data JL width). *Fix $\epsilon_\Pi, \delta_\Pi \in (0, 1)$. If $Nw \geq 2$ and*

$$n \geq n_{\text{JL}} := \frac{\log\left(2\binom{Nw}{2}/\delta_\Pi\right)}{\epsilon_\Pi^2/4 - \epsilon_\Pi^3/6}, \quad (48)$$

then, with probability at least $1 - \delta_\Pi$ over Π_n ,

$$(1 - \epsilon_\Pi) \|u - v\|_2^2 \leq \|\Pi_n u - \Pi_n v\|_2^2 \leq (1 + \epsilon_\Pi) \|u - v\|_2^2 \quad (49)$$

for every distinct $u, v \in \mathfrak{P}_N$. The statement is vacuous when $Nw < 2$.

The sufficient width is $O(\epsilon_\Pi^{-2} \log(Nw/\delta_\Pi))$ and does not depend on the ambient input dimension d . This result is deliberately orthogonal to theorem 18: it certifies the geometry of the observed raw inputs, while $\mathfrak{I}_n^\Pi$ quantifies what the projection can discard for the population prediction problem. Finite-data JL preservation is therefore not used as a surrogate for approximation quality.

7 Empirical Evaluation

The empirical study asks a deliberately small number of scientific questions with broad, structured sweeps. Minimality here refers to the number of experimental campaigns, not to sparse resource grids. We use three campaigns: (i) a representational phase diagram for the architecture $\eta = (n, R, \lambda_+)$ together with a broad controlled task atlas; (ii) a fidelity study that isolates the dependent-sample, finite-shot, and finite-window terms entering theorem 18; and (iii) a frozen-pool study that measures both the representation opportunity and the ERM selection price created by additional search budget S . All resource sweeps use dyadic grids except for qubit width: because n is a small discrete hardware resource and the width study is intended to resolve the onset of projection-information loss finely, we use the dense interval $n \in \{3, 4, \dots, 12\}$. Whenever a resource admits nested reuse, the largest realization is generated once and smaller settings are obtained by prefixes, so a sweep is not implemented as a sum of independent runs.

The main benchmark remains entirely within the formal setting of the theory. In particular, the labels are noiseless deterministic functionals of the input history, hence $\mathcal{R}^* = 0$. Observation noise and out-of-scope chaotic stress tests are therefore not used in the core empirical program: they would introduce robustness questions without identifying the role of a declared QuaRK resource.

7.1 Controlled dependent task family

Correlated uniform inputs. Fix $d \geq 1$ and $\rho_{\text{in}} \in [0, 1)$. Let $(E_t)_{t \in \mathbb{Z}}$ be i.i.d. $\mathcal{N}(0, I_d)$ and define the stationary Gaussian AR(1) process

$$G_t = \rho_{\text{in}} G_{t-1} + \sqrt{1 - \rho_{\text{in}}^2} E_t. \quad (50)$$

Then $G_t \sim \mathcal{N}(0, I_d)$ and $\text{Cov}(G_t, G_{t+q}) = \rho_{\text{in}}^q I_d$. If F_G is the standard-normal CDF, set componentwise

$$X_{t,j} := 2F_G(G_{t,j}) - 1, \quad j = 1, \dots, d. \quad (51)$$

Thus $X_t \in (-1, 1)^d \subset \mathcal{X} := [-1, 1]^d$ with uniform one-time marginals. Changing ρ_{in} changes temporal dependence without changing that marginal law; $\rho_{\text{in}} = 0$ recovers the i.i.d. drive. The componentwise transform is bijective, so it preserves the generated sigma-algebras. Stable Gaussian ARMA processes are geometrically beta mixing (Mokkadem, 1988); hence so is (X_t) for every $\rho_{\text{in}} < 1$.

For the dependence experiment it is more interpretable to parameterize ρ_{in} by its correlation half-life

$$\tau_{\text{dep}} := \frac{\log(1/2)}{\log \rho_{\text{in}}}, \quad \rho_{\text{in}} = 2^{-1/\tau_{\text{dep}}}, \quad (52)$$

for $\rho_{\text{in}} > 0$, with $\rho_{\text{in}} = 0$ treated separately as the independent reference. The dyadic grid $\tau_{\text{dep}} \in \{1, 2, 4, 8, 16\}$ therefore covers progressively more persistent dependence without abandoning a dyadic experimental design.

Legendre spatiotemporal targets. Let P_k be the ordinary Legendre polynomial of degree k , with $P_0 \equiv 1$ and $|P_k(u)| \leq 1$ on $[-1, 1]$. Fix a maximum task lag $L \geq 0$. For a nonzero multi-index $A = (a_{\ell j}) \in \mathbb{N}_0^{(L+1) \times d}$, define

$$\Psi_A(x_{-L:0}) := \prod_{\ell=0}^L \prod_{j=1}^d P_{a_{\ell j}}(x_{-\ell,j}). \quad (53)$$

We summarize the mode by its total degree, oldest active lag, and interaction arity,

$$p(A) := \sum_{\ell,j} a_{\ell j}, \quad \ell(A) := \max\{\ell : \exists j, a_{\ell j} > 0\}, \quad \iota(A) := |\{(\ell, j) : a_{\ell j} > 0\}|.$$

These quantities let the same generator vary nonlinear order, memory depth, and the number of coordinate-lag interactions.

Choose $\mathcal{A} = \{A^{(1)}, \dots, A^{(H)}\}$ and nonzero coefficients $c_1, \dots, c_H$, and let $C_c := \sum_h |c_h|$. The normalized target is

$$F_{\mathcal{A},c}^*(\mathbf{x}^-) := \frac{1}{C_c} \sum_{h=1}^H c_h \Psi_{A^{(h)}}(x_{-L:0}), \quad |F^*(\mathbf{x}^-)| \leq 1, \quad (54)$$

and the response is exactly

$$Y_t = F_{\mathcal{A},c}^*(X_{-\infty:t}). \quad (55)$$

There is no observation-noise term. The target depends only on the final $L+1$ inputs and is a continuous fading-memory functional with $\mathcal{R}^* = 0$. For $q > L$, the past of $Z_t = (X_t, Y_t)$ depends on input variables no later than time zero, whereas its future uses inputs no earlier than $q-L$. Monotonicity of absolute regularity therefore gives

$$\beta_Z(q) \leq \beta_X(q-L), \quad q > L. \quad (56)$$

The construction remains close to the information-processing framework of [Dambre et al. \(2012\)](#), but we use the Legendre modes only as controlled task generators. We do not report Dambre capacities because the original capacity interpretation assumes a linear readout and an orthogonal task basis, whereas QuaRK uses a Matérn readout and may use temporally correlated drives ([Hülser et al., 2023](#)).

7.2 Common protocol, metrics, and structured reuse

The architecture experiments use exact quantum features, $S = 1$, and an i.i.d. drive unless dependence itself is the controlled variable. This isolates representation from measurement and frozen-pool selection. Unless an axis is explicitly varied, the reference architecture and sampling gap are

$$\boldsymbol{\eta}_{\text{ref}} := (8, 16, e^{-1/32}), \quad g_{\text{ref}} := 0. \quad (57)$$

Thus a sweep changes only the resource named on that axis. We use a common reference window $w_{\text{ref}} = 512$ and a causal-reference burn-in $B = 1024$. This burned-in trajectory is treated as a certified numerical proxy, not as an exact causal state. By lemma 4, at the largest planned memory horizon $\tau_+ = 64$ its branch-state discrepancy is at most

$$2\lambda_+^B = 2e^{-16} \approx 2.25 \times 10^{-7}.$$

At readout-risk level the remaining proxy error is bounded by the same finite-window argument,

$$\varepsilon_{\text{burn}}(\boldsymbol{\eta}, B) := 4\Lambda(\Lambda + \Upsilon)L_\kappa\lambda_+^B. \quad (58)$$

Every window-fidelity result records this residual, so an observed gap is not silently attributed to finite-window initialization below the numerical causal reference precision. The finite-window campaign below verifies directly when w_{ref} is already indistinguishable from this certified proxy. Every readout uses the RMS feature metric of Equation 18, so the fixed Matérn length scale has the same interpretation as feature dimension changes. One Matérn specification and Ivanov radius are fixed before the resource sweeps and are never retuned separately for different (n, R) . This prevents either dimension-induced distance inflation or readout retuning from being misinterpreted as an architectural gain.

Because all targets are noiseless, the primary predictive metric is normalized excess mean-squared error

$$\text{NEMSE} := \frac{\widehat{\mathcal{R}}_{\text{test}} - \mathcal{R}^*}{\widehat{\text{Var}}(Y)} = \frac{\widehat{\mathcal{R}}_{\text{test}}}{\widehat{\text{Var}}(Y)}. \quad (59)$$

The normalization makes tasks with different Legendre compositions directly comparable. Theory-facing experiments additionally report the empirical shot, window, and generalization gaps defined below, while the frozen-search study reports full-sample ERM selection regret and the declared quantum acquisition run count $9NMSR$.

All predictive risks other than the explicit training-risk term in $\widehat{\Delta}_{\text{gen}}$ are evaluated on held-out windows whose labels are never used for fitting or frozen-design selection. Within a paired comparison the same held-out windows and labels are reused on both sides of a gap; “fresh measurements” means measurement randomness independent of the training cache and of the exact-feature evaluation. The independent test trajectory is also shared across competing methods within a paired repetition.

The full resource grids and task definitions are recorded in Section C. Maximal reuse is campaign-local rather than a Cartesian cache. In Campaign I a design is drawn at $R_{\text{max}} = 64$ and the setting $R \leq R_{\text{max}}$ uses the first R branches. Campaign III draws one exact-feature pool at $S_{\text{max}} = 256$ with search-branch maximum $R_{\text{search,max}} = 32$ and uses prefixes in both S and R . Campaign II, which has $S = 1$, generates each of the nine Pauli settings at $M_{\text{max}} = 8192$ repetitions and a smaller M uses the first M outcomes. The maxima are therefore never crossed into an unnecessary $R_{\text{max}}S_{\text{max}}M_{\text{max}}$ product. The same pairing principle is used for the memory endpoint: common uniforms generate the branch rates at every λ_+ while preserving the required log-uniform marginal law. The Gaussian preprocessors for the dense width sweep $n \in \{3, 4, \dots, 12\}$ are likewise coupled through prefixes of one $n_{\text{max}} = 12$ Gaussian matrix, with the $1/\sqrt{n}$ normalization restored at each width. Width-specific random mixing circuits are still drawn from their correct n -dependent laws, so this coupling pairs preprocessing randomness without treating the quantum circuits themselves as nested.

7.3 Campaign I: representational phase diagram and task atlas

This campaign tests the three representational coordinates of $\boldsymbol{\eta} = (n, R, \lambda_+)$ without finite-shot or pool-selection confounding. It contains four controlled mechanism studies and one broad generality atlas. The first three match memory, multiplexing, and width to the task difficulty they are intended to address. The fourth isolates the new mixer/observable mechanism in Section 4.5; the atlas then asks whether the resulting resource ordering survives across a wider variety of targets.

Memory endpoint versus required lag. We parameterize the upper endpoint through its longest available e-folding scale

$$\tau_+ := (-\log \lambda_+)^{-1}, \quad \lambda_+ = e^{-1/\tau_+}. \quad (60)$$

The architecture grid is $\tau_+ \in \{2, 4, 8, 16, 32, 64\}$ and the task grid is $L \in \{0, 1, 2, 4, 8, 16, 32, 64\}$, using the single-mode targets

$$F_L^*(\mathbf{x}^-) := P_2(x_{-L,1}). \quad (61)$$

At the reference values $(n, R) = (8, 16)$, the resulting $\tau_+ \times L$ surface should reveal whether longer target lags systematically require a longer reservoir-memory endpoint. The paper’s intended interpretation is challenged if the predictive optimum is essentially insensitive to L , or if larger τ_+ never helps on long lags despite the finite-window cost it incurs.

Multiplexing versus simultaneous temporal modes. To separate branch multiplicity from mere task lag, we fix $n = 8$ and $\tau_+ = 32$ and sweep

$$R \in \{1, 2, 4, 8, 16, 32, 64\}, \quad H \in \{1, 2, 4, 8\}. \quad (62)$$

For each H , the target averages H degree-two temporal modes whose lags span the dyadic memory range; the exact lag sets are specified in Section C.2. The normal QuaRK design, with independently drawn λ_r , is compared with a homogeneous-memory multiplex in which all branches share one λ . The main homogeneous

control uses the geometric center of the admissible log-rate interval. On representative slices we also scan the same dyadic τ_+ grid and select the common rate by the same full-sample empirical-risk rule used for the frozen pool, so heterogeneous memory is not compared only with an arbitrarily chosen common rate.

The central question is whether increasing R helps substantially more as H grows, and whether heterogeneous rates improve over a feature-dimension- matched homogeneous multiplex on genuinely multiscale targets. If heterogeneous and homogeneous multiplexes improve equally, the evidence supports frozen-mixing diversity but not the stronger multiscale-memory motivation. If neither improves materially beyond $R = 1$, then finite spatial multiplexing remains an approximation-feasibility mechanism but is unnecessary in the tested finite-resource regimes.

Width versus spatial information load. The width experiment fixes $R = 16$ and $\tau_+ = 32$ and uses

$$n \in \{3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}, \quad d_{\text{act}} \in \{1, 2, 4, 8, 16, 32, 64\}, \quad (63)$$

with ambient dimension $d = d_{\text{act}}$ and target

$$F_{d_{\text{act}}}^*(\mathbf{x}^-) := \frac{1}{d_{\text{act}}} \sum_{j=1}^{d_{\text{act}}} P_2(x_{0,j}). \quad (64)$$

Every additional coordinate is therefore relevant to the label. This tests whether the monotone information ordering of corollary 14 becomes predictive precisely when more input coordinates matter to the target. No strict gain is expected at every width, and the experiment is not framed as a threshold at the finite-data JL width of proposition 20; that proposition certifies observed geometry rather than population approximation.

A complementary constant-feature-budget control exploits $m_{\boldsymbol{\eta}} = 12Rn$. We compare the allocations $(n, R) \in \{(3, 16), (6, 8), (12, 4)\}$, $(n, R) \in \{(3, 32), (6, 16), (12, 8)\}$, and $(n, R) \in \{(3, 64), (6, 32), (12, 16)\}$, corresponding respectively to fixed products $Rn \in \{48, 96, 192\}$. On high-spatial-load tasks, extra branches should not fully compensate for information lost by an excessively narrow shared projection. This is the empirical counterpart of the limitation in Section 8 that increasing R cannot recover information removed by a noninjective preprocessor.

To identify whether an observed width effect is actually caused by information lost in Π_n , the same width surface includes a *projection-only control*. It receives the projected history $(\Pi_n x_{-w_{\text{ref}}+1}, \dots, \Pi_n x_0)$ directly, flattens it, and applies exactly the same RMS-normalized Matérn/Ivanov readout, with no quantum reservoir. The raw-history Matérn baseline, projection-only control, and QuaRK therefore form the ordered comparison

$$\text{raw history} \longrightarrow \text{projected history} \longrightarrow \text{QuaRK reservoir features}.$$

If performance is lost already by the projection-only control at small n and recovers with width, while QuaRK exhibits the same obstruction, the observed n -dependence is consistent with the projection-information floor rather than with a generic benefit of adding qubits. Conversely, a gap between the projection-only control and QuaRK at the same n isolates the additional finite-resource effect of the reservoir representation after preprocessing.

Mixer and observable-bank interaction order. The theoretical statement in theorem 2 is about what the frozen representation exposes before the Matérn readout. We therefore compare three dynamics at fixed (n, R, λ_+) and with paired input/preprocessor seeds:

$$\mathbf{W}_n = \mathbf{I}, \quad \mathbf{W}_n = \mathbf{V}, \quad \mathbf{W}_n = \mathbf{E}_{\tau(J_n)} \cdots \mathbf{E}_{\tau(1)} \mathbf{V}. \quad (65)$$

The middle control keeps random local rotations but suppresses every entangling edge angle, so any difference between it and the full block is attributable to graph-local operator spreading rather than to generic random single-qubit mixing. Each dynamics is evaluated once with the vertex bank $\mathcal{O}_V(n)$ and once with the complete bank $\mathcal{O}_n$. To test exactly the interaction mechanism in corollary 3, this ablation uses a small set of *preprocessor-conditioned diagnostic targets* rather than raw-coordinate atlas tasks. After drawing the shared

$\mathbf{\Pi}_n$ for a paired repetition, set $\alpha_j(x) := \vartheta((\mathbf{\Pi}_n x)_j)$ and generate

$$F_1^{\text{mix}}(\mathbf{x}^-) := \cos \alpha_1(x_0), \quad (66)$$

$$F_2^{\text{mix}}(\mathbf{x}^-) := \sin \alpha_1(x_0) \sin \alpha_2(x_0), \quad (67)$$

$$F_3^{\text{mix}}(\mathbf{x}^-) := \cos \alpha_1(x_0) \sin \alpha_2(x_0) \sin \alpha_3(x_0). \quad (68)$$

All mixer/observable variants in that repetition share the same preprocessor and therefore the same labels. The hierarchy is deliberately diagnostic: F_1^{mix} is directly one-site, F_2^{mix} is directly available to an identity mixer only through a two-site observable but is also activated from a vertex seed by generic mixing, and F_3^{mix} matches the first three-stream term proved for a mixed edge seed. Because the general 36-task atlas is generated independently of $\mathbf{\Pi}_n$, these conditional targets are not used for broad performance claims.

Alongside NEMSE, we report a task-independent Pauli-spreading diagnostic. If $\mathbf{W}_n^\dagger \mathbf{O} \mathbf{W}_n = \sum_P c_{O,P} P$, define

$$\text{Spr}(\mathbf{W}_n; \mathcal{O}) := \frac{1}{|\mathcal{O}|} \sum_{O \in \mathcal{O}} \sum_{\text{supp}(P) \not\subseteq \text{supp}(O)} |c_{O,P}|^2. \quad (69)$$

Pauli orthonormality makes this a normalized fraction of operator weight that has left the seed support. It is exactly zero for the identity and local-only controls and positive almost surely for the relevant seeds of the full mixer by theorem 2. We additionally record one fixed effective-dimension diagnostic. If F_c denotes the column-centered exact feature matrix on the common diagnostic sample, we use its stable rank

$$r_{\text{stab}}(F_c) := \frac{\|F_c\|_{\text{F}}^2}{\|F_c\|_2^2}, \quad (70)$$

with the convention $r_{\text{stab}}(0) = 0$. These diagnostics are motivated by recent REC/eigentask and Krylov-observability analyses, which distinguish the potential function space of an encoding from the independent information actually exposed by a physical reservoir (Hu et al., 2023; Ćindrak et al., 2025b;a). They are diagnostics, not tuned hyperparameters. If the full mixer increases operator spreading but neither feature rank nor interaction-task performance, the paper will report the important negative conclusion that spreading alone does not make the additional representation useful.

The theorem-aligned targets above deliberately favor interpretability over generality. To prevent the mixer claim from resting only on targets constructed from the realized encoding angles, we repeat the identical identity/local-only/full-mixer $\times$ vertex/complete-bank ablation on three *preprocessor-independent* atlas tasks: F_8^{mem} as a low-interaction control, F_{16}^{int} as a high-arity spatial interaction, and F^{C6} as a multiscale interaction task. These labels are generated before and independently of $\mathbf{\Pi}_n$. The desired conclusion is not that mixing wins uniformly: comparable performance on F_8^{mem} together with a benefit on one or both interaction-heavy tasks would be stronger evidence that the provable operator enrichment becomes useful specifically when the target requires it.

Finally, the vertex-versus-edge comparison receives one finite-shot *cost-matched* check using the full cycle mixer on the preprocessor-independent interaction task F^{C6} . Let $M_{\text{E}}^{\text{cmp}}$ be a dyadic per-setting repetition count frozen before the runs. The complete bank uses its nine settings with $M_{\text{E}}^{\text{cmp}}$ repetitions each, while the vertex bank uses the three global X/Y/Z settings with $M_{\text{V}}^{\text{cmp}} = 3M_{\text{E}}^{\text{cmp}}$ repetitions each. Both therefore use exactly $9M_{\text{E}}^{\text{cmp}}$ branch-circuit runs per input window. This separates the question “do edge observables expose useful interactions?” from the trivial fact that the complete bank normally spends more state preparations.

Broad task atlas and scientific baselines. The controlled surfaces above isolate mechanisms. To test generality without forming a Cartesian product, we additionally evaluate a fixed 36-task atlas covering dyadic memory depth, simultaneous temporal modes, active spatial dimension, nonlinear degree, interaction arity at fixed total degree, and six composite spatiotemporal targets. The explicit formulas are given in Section C.2. The atlas is evaluated at three predeclared architecture tiers,

$$\boldsymbol{\eta}_{\text{small}} = (4, 4, e^{-1/8}), \quad \boldsymbol{\eta}_{\text{medium}} = (8, 16, e^{-1/32}), \quad \boldsymbol{\eta}_{\text{large}} = (16, 64, e^{-1/64}), \quad (71)$$

rather than by tuning an architecture separately for every task.

Only two external baselines are required. A direct finite-window Matérn regressor receives the same observed raw history and asks whether any reservoir representation is needed. A frozen classical multiscale reservoir uses the same number of branches, matched heterogeneous forgetting scales, an exposed feature dimension matched to $12Rn$, and the same Matérn/Ivanov readout; it asks whether the useful temporal behavior requires the quantum representation rather than the multiscale reservoir structure alone. The purpose is not a leaderboard. In particular, simple tasks on which the direct kernel dominates are informative negative results: they identify regimes in which QuaRK’s resource expenditure is unnecessary.

7.4 Campaign II: empirical fidelity decomposition

The second campaign examines the three fidelity terms entering theorem 18. Exact simulation and the certified burned-in causal proxies of Section 7.2 let us isolate the discrepancies rather than infer them from one aggregate test error. The theoretical quantities remain sufficient upper bounds; the empirical objective is to test their qualitative resource scalings and expose their conservatism.

Finite-shot scaling. For

$$M \in \{1, 2, 4, 8, 16, 32, 64, 128, 256, 512, 1024, 2048, 4096, 8192\}, \quad (72)$$

we use the dyadic architecture sequence

$$(n, R) \in \{(4, 1), (4, 2), (4, 4), (4, 8), (8, 8), (8, 16), (16, 16), (16, 32), (16, 64)\},$$

which spans $Rn \in \{4, 8, 16, 32, 64, 128, 256, 512, 1024\}$; the memory endpoint is fixed at $\tau_+ = 32$. For every M , the readout h_M is trained on the corresponding measured training features and then evaluated on one common held-out set both with fresh grouped measurements and with exact finite-window features. The empirical risk discrepancy is

$$\hat{\Delta}_{\text{shot}}(M) := \left| \hat{\mathcal{R}}_{D,w}^{\text{op}}(h_M) - \hat{\mathcal{R}}_{D,w}(h_M) \right|. \quad (73)$$

The nine-setting measurement simulator samples complete global Pauli outcomes; all compatible observables are computed from the same shot record. It does not add independent coordinatewise noise. We plot the gap against both M and $M^{-1/2}$, stratified by the architecture sequence above. Under the RMS Matérn geometry, proposition 7 predicts that the worst-case risk sensitivity no longer grows merely because Rn increases. Approximate collapse of the architecture curves at common M would support this dimension-normalized prediction; systematic separation would quantify how state-dependent variances and representation geometry make the sufficient bound conservative in practice.

Finite-window scaling. We use

$$w \in \{1, 2, 4, 8, 16, 32, 64, 128, 256, 512\}, \quad \tau_+ \in \{2, 4, 8, 16, 32, 64\}, \quad (74)$$

and fix $(n, R) = (8, 16)$. We construct the burned-in causal proxy after the common $B = 1024$ steps. For a readout fitted at each finite w , we compare the same function on the same held-out windows using finite-window and proxy-causal exact features,

$$\hat{\Delta}_{\text{win}}(w) := \left| \hat{\mathcal{R}}_{D,\infty}(h_w) - \hat{\mathcal{R}}_{D,w}(h_w) \right|. \quad (75)$$

Here the symbol ∞ denotes the causal quantity being approximated; the implemented proxy differs from it in risk by at most $\varepsilon_{\text{burn}}(\boldsymbol{\eta}, B)$ from Equation 58. That residual is reported beside the observed gap and is therefore part of the numerical error budget rather than an unquantified burn-in assumption. The desired pattern is exponential decay in w , with visibly slower decay for larger τ_+ . This demonstrates the central memory tradeoff: longer memory can improve representation on long-lag tasks while simultaneously requiring a longer implementation window to approximate the causal reservoir.

Dependence, raw-time gap, and sample size. The dependence map uses the independent reference together with $\tau_{\text{dep}} \in \{1, 2, 4, 8, 16\}$ and

$$g \in \{0, 1, 2, 4, 8, 16, 32, 64\}. \quad (76)$$

At a fixed dyadic training size $N_0 = 1024$, this yields a dependence-by-gap surface. We then fix the strongest dependence level $\tau_{\text{dep}} = 16$ and repeat the gap sweep for

$$N \in \{128, 256, 512, 1024, 2048, 4096\}. \quad (77)$$

The reference architecture η_{ref} , a short-memory target, and exact features are used so that the experiment isolates dependent-data sampling. A long independent test realization approximates operational population risk, giving

$$\hat{\Delta}_{\text{gen}} := \left| \hat{\mathcal{R}}_{D,w}(h) - \hat{\mathcal{R}}_{D,w}^{\text{test}}(h) \right|. \quad (78)$$

The expected qualitative behavior is smaller discrepancy with larger N and, under correlated inputs, with larger raw-time gap g . We do not treat a numerical evaluation of the conservative uniform bound in proposition 9 as a separate experiment. The plots nevertheless mark the logarithmic-in- N gap scaling suggested by corollary 10; this tests whether the theorem’s explicit sampling prescription has the correct qualitative dependence even when its worst-case constants are conservative.

7.5 Campaign III: frozen-pool search and resource economics

The third campaign directly stress-tests the finite-pool mechanism in proposition 16 and theorem 17: whether larger random pools actually contain better designs, and whether the full-sample ERM rule can exploit that opportunity without its finite-sample selection price dominating. We generate one maximal pool with

$$S_{\text{max}} = 256, \quad S \in \{1, 2, 4, 8, 16, 32, 64, 128, 256\}, \quad (79)$$

and evaluate nested prefixes. The main search curves use exact features so that S is not confounded with finite-shot noise. Six representative atlas tasks are used: short single-scale memory, long single-scale memory, multiscale memory, high spatial load, high nonlinear degree, and one composite target. We fix $n = 8$ and $\tau_+ = 32$ and repeat the nested search curve at $R \in \{1, 2, 4, 8, 16, 32\}$, all obtained from the same $R_{\text{search,max}} = 32$ candidate realizations, to test whether spatial multiplexing and external random search partially substitute for one another.

For every prefix, the manuscript’s full-sample ERM rule in Equation 21 selects D_S^* . Each candidate’s readout is fitted on the training sample only and then frozen; no readout is refit on test data. We report the selected predictor’s independent-test risk together with the oracle-best of those already-fitted candidate predictors in the same prefix,

$$\hat{\mathcal{R}}_{\text{oracle}}(S) := \min_{s \leq S} \hat{\mathcal{R}}_{\text{test}}(D_s), \quad \hat{\Delta}_{\text{sel}}(S) := \hat{\mathcal{R}}_{\text{test}}(D_S^*) - \hat{\mathcal{R}}_{\text{oracle}}(S). \quad (80)$$

The oracle curve asks whether larger pools contain genuinely better frozen representations; the regret asks how much of that opportunity is lost to finite-sample ERM selection. A representative finite-shot ranking check, using dyadic prefixes up to $S = 64$ and a predeclared M fixed before inspecting the search results, verifies that the exact-feature conclusion is not an artifact of noiseless feature access.

The main S curves use the reference sample size, so we additionally include one compact sample-size-by-search slice to test the finite-sample selection price itself. At fixed $(n, R, \tau_+) = (8, 16, 32)$ and exact features, we evaluate

$$N \in \{256, 1024, 4096\}, \quad S \in \{1, 16, 64, 256\}, \quad (81)$$

on two predeclared difficult tasks, F_8^{multi} and F^{C6} . The same nested candidate pool is reused within each repetition. We report both oracle improvement and ERM selection regret. Increasing S should continue to improve the oracle opportunity, while any over-selection effect should weaken as N grows; this is the direct empirical counterpart of the theorem’s logarithmic S price inside the uniform finite-sample term. Failure

of regret to decrease with N would be evidence that the practical selector behaves more poorly than the available uniform theory suggests.

The quantum acquisition cost corresponding to any operational setting is reported under the reusable- n -qubit hardware convention as the number of independent branch-circuit executions/state preparations,

$$C_Q(N, M, S, R) = 9NMSR. \quad (82)$$

This is intentionally distinct from the resources of one execution: processing one length- w input window has logical trajectory depth $O(w)$ and uses $O(n)$ qubits under the declared hardware-reuse schedule. Fully parallel branch instantiation could trade up to the factor R in wall-clock executions for $O(Rn)$ physical width; C_Q deliberately keeps the reusable-hardware convention fixed so performance curves have one acquisition-cost definition. Because the sweep itself uses nested caches, producing all S values up to a fixed $S_{\max}$ costs the maximal pool once rather than the sum of all prefix costs. Performance is therefore shown both against S and against the state-preparation budget. If the oracle improves but ERM-selected test performance does not, the frozen design law supplies useful diversity but the finite-sample selection price is dominating. If neither curve improves beyond small S , then pool search should be presented as optional rather than as a necessary source of predictive performance.

7.6 Reporting plan

The study is summarized by four information-dense figures and one compact baseline table. Figure 1 contains the three resource response surfaces (τ_+, L) , (R, H) , and (n, d_{act}) together with the compact mixer/observable ablation, including the homogeneous-memory and constant- Rn controls. The width panel overlays the raw-history, projection-only, and QuaRK curves needed to identify the preprocessing bottleneck. The mixer panel reports NEMSE, Spr, and feature stable rank for the paired controls, with the preprocessor-independent mixer tasks shown as a small generalization slice; the finite-shot vertex/edge result is reported at matched branch-circuit run count. Figure 2 is the 36-task atlas, showing the three fixed QuaRK tiers and the two scientific baselines. Figure 3 reports the empirical shot, window, and dependent-sample gaps from Campaign II with the corresponding theoretical scalings as reference, not as fitted targets. Figure 4 reports ERM-selected and oracle pool performance against both S and $9NMSR$, together with the compact $N \times S$ selection-regret slice. The baseline table gives numerical NEMSE and uncertainty for representative short-memory, multiscale, and high-spatial-load regimes. No additional hyperparameter leaderboard is required.

8 Discussion and Limitations

QuaRK is designed to make resource choices visible rather than to argue that one quantum reservoir family dominates classical alternatives. The architecture η separates representation width, reservoir multiplicity, and memory range, while M , S , and Λ govern measurement precision, frozen-design search, and readout capacity. The final risk theorem reflects that separation: width controls an information floor, the frozen pool controls the chance of reaching near that floor, and the remaining resources control statistical, measurement, and finite-window discrepancies.

The projection-information floor is target dependent. Under the canonical Gaussian coupling it cannot increase with width, and for $n \geq d$ it vanishes almost surely because the projection is injective. This is not a claim that each additional qubit strictly improves finite-sample prediction. A target may already factor through a narrower projection. Larger n still increases physical circuit width and the number of exposed observables, but the RMS Matérn metric prevents that larger coordinate count from mechanically inflating kernel distances or the shot/window risk-sensitivity bounds. The empirical width study is therefore intended to identify tasks for which the additional projected information is actually useful.

The finite-data projection theorem answers a different question. Johnson–Lindenstrauss preservation concerns only the finite raw inputs that appear in the observed windows and can permit $n \ll d$. It does not eliminate the population information floor and is not used in the end-to-end risk proof. Conversely, the population result does not prescribe the smallest useful width for a particular finite dataset. Keeping these statements separate avoids turning finite-set geometry preservation into an unsupported performance claim.

The random-pool result also has a precise limitation. Once an architecture contains a strictly good support point, continuity and full support imply a positive target-dependent mass $\mathbf{p}_{\text{hit}}$, and the probability of missing every good design is exactly $(1 - \mathbf{p}_{\text{hit}})^S$. The paper does not provide a distribution-free numerical lower bound on $\mathbf{p}_{\text{hit}}$. Such a bound would require quantitative small-ball control for the full random circuit and memory-rate law around target-useful parameter regions and could be extremely conservative. The theorem therefore explains the role and asymptotic leverage of S without pretending that a universal finite pool size can be prescribed for every task.

The mixer theorem should likewise be read at the correct level. The identity support point remains sufficient for the asymptotic Stone–Weierstrass construction once multiplexing and a universal readout are unrestricted. What the nontrivial W_n adds is a finite-resource mechanism: generic edge gates spread seed observables to larger Pauli supports, while the cycle bounds that spreading by graph neighborhoods and the edge bank supplies two-site seeds. This does not imply that entanglement monotonically improves predictive risk, nor that the cycle is an expressivity-optimal topology. The mixer ablation therefore reports operator spreading and feature-rank diagnostics alongside task error; if spreading increases without a predictive benefit, the architectural hypothesis is rejected for that regime.

The statistical and physical fidelity bounds are sufficient and can be conservative. The finite-shot term uses coordinatewise variance bounds, the dependent-data term uses independent-block coupling and a uniform RKHS bound, and the finite-window term uses worst-case contraction. Their value is interpretability rather than numerical tightness. In particular, the condition $\delta_g > 0$ can require the raw-time gap to grow with sample size when $\beta_Z(g)$ is not negligible. This restriction is explicit rather than asymptotically hidden: under the geometric envelope $\beta_Z(q) \leq C_\beta e^{-c_\beta q}$, corollary 10 shows that $g = O(\log N)$ suffices at fixed confidence. The constants remain process dependent, so the empirical study reports observed counterparts of these gaps rather than treating the worst-case bounds as fitted predictions.

Finally, the core empirical program remains inside a controlled beta-mixing, noiseless task family, where the Bayes risk is known and resource–difficulty interactions can be isolated. Good behavior on chaotic or noisy tasks would be a robustness result, not an extension of the theorem. The resource accounting uses the number of independent state preparations/circuit runs as the acquisition metric under a reusable- n -qubit hardware convention. This should not be confused with the resources of one run: a length- w trajectory has logical depth $O(w)$ and width $O(n)$. Fully parallelizing the R branches would instead trade wall-clock executions for $O(Rn)$ width. Hardware-specific gate times, calibration drift, and fault-tolerant overhead remain outside scope. No claim of quantum advantage follows from the current theory alone.

9 Conclusion

QuaRK organizes frozen quantum reservoir learning around a compact hierarchy: an architecture $\boldsymbol{\eta}$ specifies the representation, a design $\mathbf{D} \sim \mathbb{P}_{\boldsymbol{\eta}}$ fixes one random realization, the causal map $\Phi_{\mathbf{D},\infty}$ exposes topology-local observables, and a dimension-normalized, norm-controlled Matérn readout performs regression. The RMS feature metric keeps one kernel specification meaningful across architectures without per- (n, R) retuning. A finite pool of such designs is selected by full-sample ERM, with no optimization of internal quantum parameters.

The theory first clarifies why the declared mixer is not merely decorative. With product input injection, the identity mixer leaves every raw observable coordinate confined to its seed qubits; generic graph-local mixing activates higher-support Pauli components, and edge observables expose the first nontrivial correlation layer and seed further spreading. This is a finite-resource feature statement rather than a claim of monotone benefit from entanglement, and the empirical mixer ablation is designed to test whether the additional observable structure is task useful.

The representation and learning results then connect directly. At width n , the projection-information floor isolates what the shared Gaussian preprocessing can discard; it is nonincreasing under coupled widths and vanishes almost surely in the injective regime. QuaRK’s finite-multiplex support family can approach that floor, and continuity turns a strictly good support design into positive random-design mass. Pool size S therefore has an explicit probabilistic role through the miss probability $(1 - \mathbf{p}_{\text{hit}})^S$, while the uniform

beta-mixing bound controls the overfitting price of selecting from that same pool. Adding finite-shot and finite-window fidelity yields a single excess-risk certificate for the operational predictor returned by the algorithm. For geometrically mixing processes, the accompanying gap corollary makes the dependence requirement explicit through a sufficient $O(\log N)$ raw-time gap.

A complementary finite-data JL theorem remains available for calibrating observed preprocessing geometry, but it is no longer conflated with predictive approximation. The empirical program can therefore test the theory’s intended resource roles separately: when additional width preserves useful input information, when the mixer and observable bank expose useful higher-order interactions, when spatial multiplexing and random search improve frozen representations, and when measurement or memory fidelity becomes the dominant cost. The reported acquisition metric counts branch-circuit runs under reusable n -qubit hardware; per-run width and depth remain explicit separately through n and w .

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A Technical Background and Attribution

This appendix records the standard tools imported by the proofs and separates them from the QuaRK-specific arguments.

A.1 Quantum channels and norms

An n -qubit state is a positive semidefinite trace-one operator on $(\mathbb{C}^2)^{\otimes n}$. The trace norm is $\|Q\|_1 = \text{Tr} \sqrt{Q^\dagger Q}$, and the operator norm is $\|Q\|_\infty$. Trace duality gives

$$|\text{Tr}[AB]| \leq \|A\|_\infty \|B\|_1. \quad (83)$$

The trace norm is invariant under unitary conjugation, and two density operators are at trace distance at most two.

The map $\mathcal{P}_+(Q) = \text{Tr}[Q]\rho_+$ is a replacement channel. If $\rho_+ = \sum_j r_j |j\rangle\langle j|$ and $\{|k\rangle\}_k$ is an orthonormal basis, then $L_{jk} = \sqrt{r_j} |j\rangle\langle k|$ satisfies

$$\sum_{j,k} L_{jk} Q L_{jk}^\dagger = \text{Tr}[Q]\rho_+, \quad \sum_{j,k} L_{jk}^\dagger L_{jk} = I.$$

Thus $\mathcal{P}_+$ is completely positive and trace preserving by the operator-sum representation (Kraus, 1971). The channel in Equation 13 is a convex combination of this replacement map and a unitary channel.

A.2 Gaussian finite-set embeddings

For a fixed vector $v \neq 0$ and a matrix with independent $\mathcal{N}(0, 1/n)$ entries, the ratio $\|\mathbf{\Pi}_n v\|_2^2 / \|v\|_2^2$ is a normalized chi-square variable. The standard Gaussian-projection tail bound used in the proof of proposition 20 is

$$\Pr \left[\left| \|\mathbf{\Pi}_n v\|_2^2 - \|v\|_2^2 \right| > \epsilon \|v\|_2^2 \right] \leq 2 \exp \left[-n \left(\frac{\epsilon^2}{4} - \frac{\epsilon^3}{6} \right) \right], \quad \epsilon \in (0, 1).$$

The manuscript uses this only as a finite-set statement; it does not claim that a projection into $\mathbb{R}^n$ is globally injective when $n < d$.

A.3 Beta mixing and unit-block coupling

Absolute regularity is monotone under sigma-algebra inclusion, and measurable maps cannot increase it (Bradley, 2005). These properties yield Equation 27 from the raw-time gap and independent measurement seeds.

The independent-block technique of Yu (1994) compares dependent collections with independent blocks having the same marginals. The general Rademacher analysis of Mohri & Rostamizadeh (2008) allows arbitrary block lengths. This paper fixes that length to one. The $N = 2\mu$ windows are split into odd and even collections of μ single-window blocks, so the coupling cost is controlled by $\beta_Z(g)$.

For independent observations, a class bounded by $L_{\max}$ obeys

$$\sup_{f \in \mathcal{F}} \left| \mathbb{E}f - \widehat{\mathbb{E}}f \right| \leq 2\mathfrak{R}_\mu(\mathcal{F}) + 3L_{\max} \sqrt{\frac{\log(2/\alpha)}{2\mu}} \quad (84)$$

with probability at least $1 - \alpha$ under the convention used in the unit-block Rademacher bound. The contraction principle transfers the prediction-class complexity to squared loss (Bartlett & Mendelson, 2002).

A.4 Grouped Pauli measurements

A global Pauli setting assigns one of X , Y , and Z to each qubit. One shot returns a sign for every qubit, and products of those signs estimate every compatible Pauli string. Because QuaRK declares its graph-local observables in advance, a deterministic cover can replace a generic shadow ensemble. Overlapping tomography relates such covers to covering arrays and provides constant-size constructions for nearest-neighbor two-body Pauli marginals (Hansen et al., 2025).

For the cycle, a balanced nine-setting cover can be constructed explicitly. Fix a proper vertex coloring $c : \mathcal{V}_n \rightarrow \{0, 1, 2\}$ and identify $\{X, Y, Z\}$ with $\{0, 1, 2\}$. For $(a, b) \in \{0, 1, 2\}^2$, define the setting whose basis at vertex v is $a + bc(v)$ modulo 3. Every one-qubit Pauli basis then appears in exactly three settings at each vertex. For adjacent vertices u, v , we have $c(u) \neq c(v)$, so

$$(a, b) \mapsto (a + bc(u), a + bc(v)) \pmod{3}$$

is a bijection of $\{0, 1, 2\}^2$. Hence every two-qubit Pauli pair on an edge appears in exactly one of the nine settings. This gives $a_n(O) = 3$ for $O \in \mathcal{O}_V(n)$ and $a_n(O) = 1$ for $O \in \mathcal{O}_E(n)$.

If $\phi = \text{Tr}[O\rho]$, every compatible projective measurement of O produces a sign with mean ϕ and variance $1 - \phi^2$. Pooling all $a_n(O)M$ compatible independent state preparations therefore gives coordinate variance

$$\frac{1 - \phi^2}{a_n(O)M}.$$

The same shot may contribute to several compatible coordinate estimators, so those estimators need not be independent. The finite-shot proof only sums their coordinatewise mean-squared errors and therefore does not require such independence.

A.5 Expressivity and observability provenance

The mixer analysis in Section 4.5 uses only elementary Pauli conjugation and real-analyticity; it does not require a capacity theorem from the literature. Its interpretation is informed by two complementary strands. For rotation-based quantum models, the data-encoding generators determine the available Fourier frequencies, whereas later circuit structure controls which Fourier coefficients can be populated (Schuld et al., 2021). Recent QRC work extends this frequency/capacity viewpoint to temporal settings and shows explicitly that

increasing reservoir size or dynamical complexity need not increase resolvable expressive capacity when input injection is the limiting factor (Schütte et al., 2026).

A separate line studies what the measurement interface can resolve. The resolvable expressive capacity/eigentask framework incorporates finite sampling noise (Hu et al., 2023), while Krylov observability uses time-evolved measured operators to estimate the effective observable space and has been shown to track QRC information-processing capacity closely (Čindrak et al., 2025b;a). QuaRK therefore uses Pauli support growth and the observable-bank ablation as transparent mechanism diagnostics. We do not identify Pauli weight, entanglement, Krylov dimension, or expressive capacity with predictive advantage without the corresponding task evidence.

A.6 Universality provenance

Classical reservoir universality is commonly proved through fading memory, point separation, and Stone–Weierstrass algebra arguments (Grigoryeva & Ortega, 2018). Monzani & Prati (2025) give a unified classical/quantum formulation and identify spatial multiplexing—parallel, uncoupled reservoirs driven by the same input with a joint readout—as a QRC mechanism for constructing the required polynomial algebra. We use that established mechanism only inside the width-resolved approximation argument in Section 6.4; no novelty is claimed for the universality principle itself.

The Matérn RKHS plays a different role. Its full-support radial spectral density makes it universal on compact Euclidean subsets (Sriperumbudur et al., 2011), while normalization gives an RKHS ball of radius Λ the standard Rademacher bound of order $\Lambda/\sqrt{\mu}$. QuaRK additionally evaluates the kernel on the RMS feature metric $\|z - z'\|_2/\sqrt{p}$. For each finite p this is simply the invertible coordinate rescaling $z \mapsto z/\sqrt{p}$, so strict positive definiteness and universality are unchanged; the section-map Lipschitz constant with respect to the unscaled Euclidean feature vector is reduced by $1/\sqrt{p}$. We therefore do not require the fixed-radius Matérn ball itself to be a polynomial algebra. The polynomial algebra appears only as an intermediate spatial-multiplexing construction and is approximated on the resulting compact feature image by the Matérn RKHS. This separation is also motivated by the QRC risk analysis of Chmielewski et al. (2025), who show unfavourable qubit scaling for the polynomial-readout classes they study and discuss spatial multiplexing as an alternative algebraic mechanism.

A.7 Novelty boundary

The paper does not claim novelty for Gaussian random projection, replacement channels, beta-mixing coupling, Rademacher complexity, Ivanov regularization, Matérn kernels, covering arrays, or reservoir model selection in general. The QuaRK-specific contribution is the topology-aligned random reservoir family, the graph-local finite-resource spreading statement for that declared mixer/observable architecture, and explicit resource-to-risk-fidelity guarantees under dependent temporal sampling and grouped quantum measurement.

B Proofs

B.1 Balanced circuit

For the matching-order permutation τ contained in Γ , the unitary in Equation 11 is $\mathbf{W}_n = \mathbf{E}_{\tau(J_n)} \cdots \mathbf{E}_{\tau(1)} \mathbf{V}$. Every vertex and edge appears once. Setting all gate angles to zero gives the identity, and each matrix entry is real analytic in the finitely many angles.

B.2 Mixer locality and operator spreading

Proof of proposition 1. Let $A = \text{supp}(O)$. When $\mathbf{W}_n = \mathbf{I}$, the input unitary factors as $\mathbf{U}_n^{\text{in}}(x) = \mathbf{U}_A^{\text{in}}(x_A) \otimes \mathbf{U}_{A^c}^{\text{in}}(x_{A^c})$, where x_A denotes the projected angle coordinates indexed by A . Taking the partial trace of Equation (13) over A^c therefore gives the closed reduced recursion

$$\rho_A \longmapsto \lambda \mathbf{U}_A^{\text{in}}(x_A) \rho_A \mathbf{U}_A^{\text{in}}(x_A)^\dagger + (1 - \lambda) \rho_{+,A}.$$

Hence the reduced state on A , and therefore $\text{Tr}[\mathcal{O}\rho_A]$, is independent of every projected input coordinate outside A at every finite time. Passing to the unique causal limit preserves the same dependence restriction. The claimed one- and two-stream conclusions follow from $|A| = 1$ for $\mathcal{O}_V(n)$ and $|A| = 2$ for $\mathcal{O}_E(n)$. $\square$

Proof of theorem 2. Conjugation by the vertex layer $\mathbf{V}$ cannot enlarge Pauli support because it is a tensor product of one-qubit unitaries. Consider one matching layer. An edge gate on $e = \{a, b\}$ commutes with any Pauli component whose support is disjoint from e ; if the component intersects e , conjugation can introduce the other endpoint but no site outside e . Since the gates in a matching act on disjoint edges, one matching layer expands a support set by at most one graph neighborhood. Iterating through the J_n matching layers gives $\mathcal{N}_{C_n}^{J_n}(A)$, and repeating the same argument for s blocks gives $\mathcal{N}_{C_n}^{sJ_n}(A)$.

It remains to show that spreading is not merely allowed by this light cone but is generic under the declared law. Fix an oriented edge $u \rightarrow v$ and define the Pauli coefficient

$$c_{uv}(\omega) := 2^{-n} \text{Tr}[\mathbf{Z}_u \mathbf{Z}_v \mathbf{W}_n(\omega)^\dagger \mathbf{X}_u \mathbf{W}_n(\omega)].$$

For every fixed matching permutation this coefficient is a real-analytic function of the gate angles and, in local analytic charts, of the generator axes. It is not identically zero. Indeed, set every gate angle to zero except the edge $\{u, v\}$, choose its generator $\mathbf{G} = \mathbf{Y}_u \mathbf{Z}_v$, and take an interior angle $\theta \in (0, \gamma)$. Since $\{\mathbf{G}, \mathbf{X}_u\} = 0$ and $\mathbf{G}^2 = \mathbf{I}$,

$$e^{i\theta \mathbf{G}/2} \mathbf{X}_u e^{-i\theta \mathbf{G}/2} = \mathbf{X}_u \cos \theta + \mathbf{Z}_u \mathbf{Z}_v \sin \theta,$$

so $c_{uv} = \sin \theta \neq 0$. A nonzero real-analytic function has a measure-zero zero set. The axis law is smooth with full support on the sphere and the angle law is absolutely continuous with positive density on its interior, hence $c_{uv} \neq 0$ almost surely.

For a length-two path $u - v - w$, define analogously

$$d_{uvw}(\omega) := 2^{-n} \text{Tr}[\mathbf{X}_u \mathbf{Z}_v \mathbf{Z}_w \mathbf{W}_n(\omega)^\dagger (\mathbf{X}_u \mathbf{X}_v) \mathbf{W}_n(\omega)].$$

Set all angles to zero except the edge $\{v, w\}$, choose its generator $\mathbf{Y}_v \mathbf{Z}_w$, and again take an interior nonzero angle. The same anticommuting-Pauli identity gives

$$e^{i\theta \mathbf{Y}_v \mathbf{Z}_w/2} (\mathbf{X}_u \mathbf{X}_v) e^{-i\theta \mathbf{Y}_v \mathbf{Z}_w/2} = \mathbf{X}_u \mathbf{X}_v \cos \theta + \mathbf{X}_u \mathbf{Z}_v \mathbf{Z}_w \sin \theta.$$

Thus d_{uvw} is also a nontrivial real-analytic coefficient and is nonzero almost surely under the same law. $\square$

Proof of corollary 3. For one input step from ρ_+ , the unitary part of a feature is

$$\text{Tr}[\rho_+ \mathbf{U}_n^{\text{in}}(x)^\dagger \mathbf{W}_n^\dagger \mathbf{O} \mathbf{W}_n \mathbf{U}_n^{\text{in}}(x)].$$

For the encoded one-qubit $|+\rangle$ state,

$$\langle \mathbf{X}_j \rangle = \cos \alpha_j(x), \quad \langle \mathbf{Z}_j \rangle = -\sin \alpha_j(x), \quad \langle \mathbf{Y}_j \rangle = 0.$$

The nonzero $\mathbf{Z}_u \mathbf{Z}_v$ coefficient from theorem 2 therefore contributes a nonzero multiple of $\sin \alpha_u \sin \alpha_v$ to the $\mathbf{X}_u$ feature. The nonzero $\mathbf{X}_u \mathbf{Z}_v \mathbf{Z}_w$ coefficient similarly contributes a nonzero multiple of $\cos \alpha_u \sin \alpha_v \sin \alpha_w$ to the $\mathbf{X}_u \mathbf{X}_v$ feature. These tensor-product trigonometric monomials correspond to unique $X/Z/I$ Pauli strings, so their coefficients cannot be canceled by a different Pauli component. Multiplication by the branch factor $\lambda > 0$ preserves them, while the replacement part adds only the fixed ρ_+ expectation. The identity-mixer comparison follows from proposition 1. $\square$

B.3 Contraction and finite-window control

Proof of lemma 4. The replacement terms cancel on state differences. The remaining term is λ times unitary conjugation, which preserves the trace norm. Iterating two trajectories driven by the same final w inputs produces the factor λ^w , and the trace distance between two states is at most two. Strict contraction also gives existence and uniqueness of the causal state. $\square$

Lemma 21 (Matérn section stability). *For every feature dimension p ,*

$$\|\kappa_p(z, \cdot) - \kappa_p(z', \cdot)\|_{\mathcal{H}_p} \leq L_\kappa d_p^{\text{rms}}(z, z') = \frac{L_\kappa}{\sqrt{p}} \|z - z'\|_2. \quad (85)$$

Proof. The reproducing property gives $\|\kappa_p(z, \cdot) - \kappa_p(z', \cdot)\|_{\mathcal{H}_p}^2 = 2(1 - \kappa_p(z, z'))$. Relative to the raw feature coordinate, the RMS-rescaled kernel is the ordinary Matérn kernel applied to $z/\sqrt{p}$; its normalized spectral law therefore has covariance $\nu I/(p(\nu - 1)\xi^2)$. Applying $1 - \cos u \leq u^2/2$ yields the claim. $\square$

Proof of proposition 5. Apply lemma 4 to all R branches. Since every $O \in \mathcal{O}_n$ has operator norm one, trace duality gives, uniformly in the input history,

$$\|\Phi_{D,\infty}(\mathbf{x}^-) - \Phi_D(x_{-w+1:0})\|_2 \leq 2\sqrt{q_n \sum_{r=1}^R \lambda_r^{2w}}.$$

For $h \in \mathcal{H}_{m_\eta}^\Lambda$, lemma 21 and the reproducing property multiply this by $\Lambda L_\kappa/\sqrt{Rq_n}$. Thus the raw feature-count factor cancels and the prediction discrepancy is bounded by $2\Lambda L_\kappa \sqrt{R^{-1} \sum_r \lambda_r^{2w}}$. Finally, $|(u-y)^2 - (v-y)^2| \leq 2(\Lambda + \Upsilon)|u - v|$. Since $\lambda_r \leq \lambda_+$ for every branch, Equation 23 follows. $\square$

Proof of corollary 6. Solve the upper bound in Equation 23 for λ_+ under $\varepsilon_{\text{win}}(\boldsymbol{\eta}) \leq \epsilon_W$. $\square$

B.4 Grouped-measurement control

Proof of proposition 7. For branch r and observable O , each compatible Pauli measurement is a sign with mean $\phi_{r,O}$ and variance $1 - \phi_{r,O}^2$. Pooling $a_n(O)M$ compatible independent preparations therefore gives coordinate variance $(1 - \phi_{r,O}^2)/(a_n(O)M)$. Summing coordinatewise mean-square errors over all branches does not require independence between distinct feature coordinates. Jensen’s inequality, lemma 21, and the bounded squared-loss inequality introduce the factor $1/\sqrt{Rq_n}$ and give the first line of Equation 25; bounding each variance by one gives the second. For the cycle cover, $\sum_{O \in \mathcal{O}_n} a_n(O)^{-1} = 10n$ and $q_n = 12n$, hence their normalized ratio is $5/6$. $\square$

Proof of corollary 8. Insert the cycle value $\varepsilon_{\text{shot}}(\boldsymbol{\eta}, M) = 2\Lambda(\Lambda + \Upsilon)L_\kappa \sqrt{5/(6M)}$ and solve $\varepsilon_{\text{shot}}(\boldsymbol{\eta}, M) \leq \epsilon_M$ for M . $\square$

B.5 Mixing and the uniform generalization bound

Lemma 22 (Augmented-window mixing). *Conditional on a realized shared projection and frozen pool, $\beta_U(1) = \beta_W(1) \leq \beta_Z(g)$.*

Proof. The raw variables supporting adjacent windows are separated by g , so monotonicity of absolute regularity gives $\beta_W(1) \leq \beta_Z(g)$. Conditional on the projection and pool, each measurement record is generated from its own window through an independent measurement kernel. Such kernels cannot increase total variation, while projection from U_j back to W_j gives the reverse inequality. $\square$

Lemma 23 (Independent measured-window deviation). *For μ independent augmented windows with the stationary one-window marginal, a fixed design, and $\alpha \in (0, 1)$, with probability at least $1 - \alpha$,*

$$\sup_{h \in \mathcal{H}_{m_\eta}^\Lambda} \left| \mathcal{R}_{D,w}^{\text{op}}(h) - \frac{1}{\mu} \sum_{j=1}^{\mu} (h(\widehat{\Phi}_D(U_j^X; U_j^M)) - U_j^Y)^2 \right| \leq \frac{4\Lambda(\Lambda + \Upsilon)}{\sqrt{\mu}} + 3(\Lambda + \Upsilon)^2 \sqrt{\frac{\log(2/\alpha)}{2\mu}}. \quad (86)$$

Proof. Kernel normalization gives prediction-class Rademacher complexity at most $\Lambda/\sqrt{\mu}$. For fixed $y \in [-\Upsilon, \Upsilon]$, the centered squared loss is $2(\Lambda + \Upsilon)$ -Lipschitz on the prediction range. The contraction principle and the standard two-sided Rademacher inequality (Bartlett & Mendelson, 2002) give the result. $\square$

Proof of proposition 9. Split the $N = 2\mu$ augmented windows into odd and even collections. The independent-block coupling of Yu (1994) replaces either collection by an independent sample with the same one-window marginal, with transfer cost at most $2(\mu - 1)\beta_Z(g)$ per parity under the convention of Mohri & Rostamizadeh (2008). For each candidate and parity, apply lemma 23 with $\alpha = \delta_g/(2S)$. A union bound over both parities and all S candidates costs δ_g , while the coupling costs $4(\mu - 1)\beta_Z(g)$. This gives Equation 28 and Equation 29. $\square$

Proof of corollary 10. Under Equation 31,

$$4(\mu - 1)\beta_Z(g) \leq 4(\mu - 1)C_\beta e^{-c_\beta g} \leq \frac{\delta}{2}.$$

Therefore $\delta_g \geq \delta/2$. Substituting this inequality into $\log(4S/\delta_g)$ gives $\log(4S/\delta_g) \leq \log(8S/\delta)$, which yields Equation 32. Since $\mu = N/2$, the sufficient gap in Equation 31 is $O(\log N)$ for fixed $(C_\beta, c_\beta, \delta)$. $\square$

Proof of corollary 11. With $a = 4/\sqrt{\mu}$ and $b = 3\sqrt{\log(4S/\delta_g)/(2\mu)}$, expanding Equation 28 gives $\varepsilon_{\text{gen}} = (a + b)\Lambda^2 + (a + 2b)\Upsilon\Lambda + b\Upsilon^2$. This expression is strictly increasing for $\Lambda \geq 0$. Solving $\varepsilon_{\text{gen}} = \epsilon_G$ by the quadratic formula gives Equation 33, and a positive solution exists exactly when $\epsilon_G > b\Upsilon^2$. $\square$

B.6 Width-resolved representation

Proof of lemma 12. Write a projected history as $\mathbf{z}^- = (\dots, z_{-1}, z_0) \in (\Pi_n \mathcal{X})^{\mathbb{Z}_-}$. Strict contraction gives a unique causal state for every admissible branch. For a fixed branch with rate $\lambda < 1$, projected inputs more than L steps in the past affect the current state by at most $2\lambda^L$ in trace norm. After choosing L so that this tail is arbitrarily small, continuity of the finite composition of the input-dependent unitary channels controls the remaining recent projected inputs. Trace duality therefore shows that every causal coordinate is continuous in the fading-memory topology on the projected history space.

For separation, take $\mathbf{z}^- \neq \mathbf{z}'^-$ and let ℓ_0 be their most recent differing lag. Some coordinate j satisfies $z_{-\ell_0, j} \neq z'_{-\ell_0, j}$. Choose the zero-angle support point $\mathbf{W}_n = \mathbf{I}$ and, only as a proof device combining two coordinates of the same observable bank, define $F_{j, \mathbf{z}^-}(\lambda) := \langle \mathbf{X}_j - i\mathbf{Z}_j \rangle$. With

$$\alpha_{j, \ell}(\mathbf{z}^-) := \vartheta(z_{-\ell, j}), \quad z_{j, s}^\circ(\mathbf{z}^-) := \exp\left(i \sum_{\ell=0}^{s-1} \alpha_{j, \ell}(\mathbf{z}^-)\right), \quad z_{j, 0}^\circ := 1,$$

the causal reset expansion is

$$F_{j, \mathbf{z}^-}(\lambda) = 1 + \sum_{s=1}^{\infty} \lambda^s (z_{j, s}^\circ(\mathbf{z}^-) - z_{j, s-1}^\circ(\mathbf{z}^-)). \quad (87)$$

The series is analytic for $\lambda \in (0, 1)$. In the difference between the two histories, all coefficients before order $\ell_0 + 1$ vanish, whereas the coefficient at that order is nonzero. Indeed, all more recent projected inputs agree and $\vartheta : \mathbb{R} \rightarrow (-\pi, \pi)$ is injective, so the first differing cumulative phase cannot differ by a nonzero multiple of 2π . The analytic difference is therefore not identically zero, and its zero set cannot contain (λ_0, λ_+) . Some rate in the support consequently makes $F_{j, \mathbf{z}^-} \neq F_{j, \mathbf{z}'^-}$. At least one of the $\mathbf{X}_j$ or $\mathbf{Z}_j$ expectations must then differ, and both belong to $\mathcal{O}_n$. $\square$

Proof of theorem 13. By lemma 12, as admissible single-reservoir designs vary, their scalar causal coordinates are continuous fading-memory functionals on $(\Pi_n \mathcal{X})^{\mathbb{Z}_-}$ and separate projected histories. The real algebra generated by these coordinates and constants is therefore dense in $\mathcal{F}_{\text{FMP}}(\Pi_n \mathcal{X})$ by the Stone–Weierstrass mechanism of Monzani & Prati (2025). Any approximating polynomial uses finitely many coordinates, and spatial multiplexing places the corresponding single-reservoir designs into one finite- R product-support design. The image of the resulting feature map is compact, so universality of the normalized Matérn kernel on compact Euclidean subsets (Sriperumbudur et al., 2011) replaces the intermediate polynomial by a finite-norm RKHS readout. Consequently the support-closure QuaRK predictors are uniformly dense in $\mathcal{F}_n^\Pi$.

Every causal QuaRK predictor is measurable with respect to $\sigma(\mathbf{\Pi}_n X_t : t \leq 0)$, because the entire driven channel sees x_t only through $\mathbf{\Pi}_n x_t$. Orthogonality of conditional expectation under squared loss therefore gives the information-theoretic lower bound $\mathfrak{J}_n^\Pi$ in Equation 34; no predictor that only sees the projected history can do better.

For the converse, let $\bar{F}_n := \mathbb{E}[F^*(X_{-\infty:0}) \mid \sigma(\mathbf{\Pi}_n X_t : t \leq 0)]$, viewed as an L^2 function of the projected history. The projected history space is compact metrizable, so continuous functions are dense in its L^2 law. Since our fading-memory class is exactly the continuous class for the product topology, for any $\xi > 0$ there exists $F_\xi \in \mathcal{F}_n^\Pi$ whose squared error relative to F^* is at most $\mathfrak{J}_n^\Pi + \xi$; this is Equation 35. Uniform density of QuaRK in $\mathcal{F}_n^\Pi$ then supplies a finite R , a support design, and a finite-norm readout arbitrarily close to F_ξ on the compact history space. Since F_ξ and F^* are bounded, the corresponding squared error converges to that of F_ξ . Thus the infimum over finite-norm QuaRK predictors is at most $\mathfrak{J}_n^\Pi + \xi$. Letting $\xi \downarrow 0$ proves equality. The radius-truncated infimum is nonincreasing in Λ , and its infimum over all finite radii is the unrestricted finite-norm infimum just identified; therefore its limit is $\mathfrak{J}_n^\Pi$. $\square$

Proof of corollary 14. From Equation 9, $\sigma(\mathbf{\Pi}_n X_t : t \leq 0)$ is contained in $\sigma(\mathbf{\Pi}_{n+1} X_t : t \leq 0)$. Conditional expectation is the orthogonal projection in L^2 , so enlarging the conditioning sigma-field cannot increase the residual squared norm. This gives $\mathfrak{J}_{n+1}^\Pi \leq \mathfrak{J}_n^\Pi$ directly.

If $n \geq d$, the Gaussian matrix $\mathbf{\Pi}_n$ has full column rank almost surely. The full input history is then measurable with respect to the projected history (indeed the inverse is linear on the image), so $\sigma(\mathbf{\Pi}_n X_t : t \leq 0) = \sigma(X_t : t \leq 0)$. Because F^* is measurable with respect to the latter sigma-field, the conditional residual in Equation 34 is zero. $\square$

B.7 Positive mass and finite-pool coverage

Proof of proposition 15. Choose $h_0 \in \mathcal{H}_{m_\eta}^\Lambda$ such that

$$\mathcal{R}_{D_0, \infty}(h_0) - \mathcal{R}^* < \mathfrak{J}_n^\Pi + \varepsilon_{\text{rep}};$$

such an h_0 exists because Equation 40 is a strict inequality. It is enough to show that the causal risk of this fixed readout is continuous in the frozen design near D_0 .

For branch r , let

$$c_r(D, D_0) := \sup_{x \in \mathcal{X}} \sup_{\rho} \|\mathcal{T}_{r,x}^D(\rho) - \mathcal{T}_{r,x}^{D_0}(\rho)\|_1.$$

The one-step channel is continuous in the finite-dimensional rate, axis, and angle parameters. In a sufficiently small neighborhood of D_0 , all rates are bounded by a common $\bar{\lambda} < 1$. The standard perturbation comparison for contractive recursions then gives

$$\sup_{\mathbf{x}^-} \|\rho_{r,\infty}^D(\mathbf{x}^-) - \rho_{r,\infty}^{D_0}(\mathbf{x}^-)\|_1 \leq \frac{c_r(D, D_0)}{1 - \bar{\lambda}}.$$

Thus the causal feature maps converge uniformly as $D \rightarrow D_0$. By lemma 21 and the reproducing property, $h_0 \circ \Phi_{D,\infty}$ also converges uniformly, and bounded squared loss makes $\mathcal{R}_{D,\infty}(h_0)$ continuous. The strict inequality therefore persists on some open neighborhood $\mathcal{N}$ of D_0 . Because $D_0 \in \text{supp}(\mathbb{P}_\eta)$, $\mathbb{P}_\eta(\mathcal{N}) > 0$, and every design in $\mathcal{N}$ is counted by Equation 39. Hence the good-design mass is positive.

For the final statement, theorem 13 gives, for any $\varepsilon_{\text{rep}} > 0$, finite R , finite Λ , and a product-support design with approximation error strictly below $\mathfrak{J}_n^\Pi + \varepsilon_{\text{rep}}$; apply the first part. $\square$

Proof of proposition 16. Conditional on $\mathbf{\Pi}_n$, the events $\{\varepsilon_{\text{app}}(D_s; \Lambda) > \mathfrak{J}_n^\Pi + \varepsilon_{\text{rep}}\}$ are independent and each has probability $1 - \mathfrak{p}_{\text{hit}}$. Their intersection therefore has probability $(1 - \mathfrak{p}_{\text{hit}})^S$, proving Equation 41. $\square$

B.8 ERM oracle and resource-to-performance guarantee

Proof of theorem 17. Work on the simultaneous event of proposition 9. For any candidate s and any $h \in \mathcal{H}_{m_\eta}^\Lambda$, the joint ERM property Equation 21 gives

$$\begin{aligned}\mathcal{R}_{\mathbf{D}^\star, w}^{\text{op}}(h^\star) &\leq \widehat{\mathcal{R}}_{\mathbf{D}^\star, w}(h^\star) + \varepsilon_{\text{gen}} \\ &\leq \widehat{\mathcal{R}}_{\mathbf{D}_s, w}(h) + \varepsilon_{\text{gen}} \\ &\leq \mathcal{R}_{\mathbf{D}_s, w}^{\text{op}}(h) + 2\varepsilon_{\text{gen}},\end{aligned}$$

where $\varepsilon_{\text{gen}} = \varepsilon_{\text{gen}}(\Lambda, S, \delta)$. Taking first the infimum over h and then the minimum over s proves Equation 42. $\square$

Proof of theorem 18. Let E_{pool} be the event that at least one sampled design satisfies $\varepsilon_{\text{app}}(\mathbf{D}_s; \Lambda) \leq \mathfrak{J}_n^\Pi + \varepsilon_{\text{rep}}$. By proposition 16, conditional on $\mathbf{\Pi}_n$,

$$\mathbb{P}(E_{\text{pool}}^c \mid \mathbf{\Pi}_n) = (1 - \mathfrak{p}_{\text{hit}})^S.$$

On E_{pool} , choose such a candidate s_0 . For any $h \in \mathcal{H}_{m_\eta}^\Lambda$,

$$\mathcal{R}_{\mathbf{D}_{s_0}, w}^{\text{op}}(h) \leq \mathcal{R}_{\mathbf{D}_{s_0}, \infty}(h) + \varepsilon_{\text{shot}}(\boldsymbol{\eta}, M) + \varepsilon_{\text{win}}(\boldsymbol{\eta})$$

by propositions 5 and 7. Taking the infimum in h gives

$$\inf_h \mathcal{R}_{\mathbf{D}_{s_0}, w}^{\text{op}}(h) - \mathcal{R}^\star \leq \mathfrak{J}_n^\Pi + \varepsilon_{\text{rep}} + \varepsilon_{\text{shot}} + \varepsilon_{\text{win}}.$$

Intersect this pool event with the event of theorem 17. A union bound gives confidence Equation 43, and substitution yields Equation 44.

For the causal exact-feature counterpart, apply propositions 5 and 7 once more to the selected readout: $\mathcal{R}_{\mathbf{D}^\star, \infty}(h^\star) \leq \mathcal{R}_{\mathbf{D}^\star, w}^{\text{op}}(h^\star) + \varepsilon_{\text{shot}} + \varepsilon_{\text{win}}$. Combining with Equation 44 proves Equation 45. $\square$

Proof of corollary 19. The first condition in Equation 46 bounds the two statistical deviations in Equation 44 by $2\epsilon_G$; the remaining two conditions are exactly corollaries 6 and 8. Substitution into theorem 18 yields Equation 47 with confidence $1 - \delta - (1 - \mathfrak{p}_{\text{hit}})^S$. $\square$

B.9 Finite-data Gaussian geometry

Proof of proposition 20. Condition on the realized input windows. For fixed $v \neq 0$, the normalized Gaussian projection satisfies

$$\Pr \left[\left| \|\mathbf{\Pi}_n v\|_2^2 - \|v\|_2^2 \right| > \epsilon_\Pi \|v\|_2^2 \right] \leq 2 \exp \left[-n \left(\frac{\epsilon_\Pi^2}{4} - \frac{\epsilon_\Pi^3}{6} \right) \right].$$

There are at most $\binom{Nw}{2}$ nonzero pairwise differences among the raw vectors appearing in the sample. A union bound under Equation 48 gives failure probability at most δ_Π . Since $\mathbf{\Pi}_n$ is independent of the temporal sample, the same probability statement holds after averaging over that sample. $\square$

C Empirical Materials

This appendix fixes the experimental objects that are part of the empirical protocol but would interrupt the main narrative: structured resource grids, exact task formulas, maximal-cache reuse, paired randomization, baseline matching, and reported diagnostics. No numerical result is inferred from this appendix; the final release will additionally contain the realized random seeds, measurement caches, fit outputs, and scripts producing every figure and table. The source project also carries a theorem-to-code implementation contract that freezes indexing, initialization, measurement, readout, selection, and paired comparison semantics before numerical results are generated.

C.1 Resource grids and fixed reference settings

All one-dimensional resource sweeps use powers of two (with zero added only where it is a meaningful boundary value), except for qubit width. Width is swept densely over the small hardware-relevant interval $3 \leq n \leq 12$ so that the empirical onset of projection-information loss is not hidden between widely spaced dyadic widths. The planned grids are

$$\begin{aligned}
\mathcal{G}_L &= \{0, 1, 2, 4, 8, 16, 32, 64\}, \\
\mathcal{G}_{\tau_+} &= \{2, 4, 8, 16, 32, 64\}, \\
\mathcal{G}_R &= \{1, 2, 4, 8, 16, 32, 64\}, \\
\mathcal{G}_H &= \{1, 2, 4, 8\}, \\
\mathcal{G}_d &= \{1, 2, 4, 8, 16, 32, 64\}, \\
\mathcal{G}_n &= \{3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}, \\
\mathcal{G}_p &= \{1, 2, 4, 8, 16, 32\}, \\
\mathcal{G}_t &= \{1, 2, 4, 8, 16\}, \\
\mathcal{G}_M &= \{1, 2, 4, 8, 16, 32, 64, 128, 256, 512, 1024, 2048, 4096, 8192\}, \\
\mathcal{G}_w &= \{1, 2, 4, 8, 16, 32, 64, 128, 256, 512\}, \\
\mathcal{G}_g &= \{0, 1, 2, 4, 8, 16, 32, 64\}, \\
\mathcal{G}_N &= \{128, 256, 512, 1024, 2048, 4096\}, \\
\mathcal{G}_S &= \{1, 2, 4, 8, 16, 32, 64, 128, 256\}, \\
\mathcal{G}_{\tau_{\text{dep}}} &= \{1, 2, 4, 8, 16\}.
\end{aligned}$$

The independent input process $\rho_{\text{in}} = 0$ is included separately from $\mathcal{G}_{\tau_{\text{dep}}}$. For a positive dependence half-life, $\rho_{\text{in}} = 2^{-1/\tau_{\text{dep}}}$. The memory endpoint is $\lambda_+ = e^{-1/\tau_+}$, so every value in $\mathcal{G}_{\tau_+}$ lies strictly above the lower endpoint $\lambda_0 = e^{-1}$.

Unless a campaign varies the corresponding object, the reference settings are $\boldsymbol{\eta}_{\text{ref}} = (8, 16, e^{-1/32})$, $N_0 = 1024$, $w_{\text{ref}} = 512$, $g_{\text{ref}} = 0$, causal-proxy burn-in $B = 1024$, $S = 1$, exact quantum features, and an i.i.d. input drive. At the largest planned memory endpoint $\tau_+ = 64$, the worst-case branch-state discrepancy between this burned-in proxy and the infinite-past causal state is at most $2e^{-16} \approx 2.25 \times 10^{-7}$; the corresponding risk residual is the quantity $\varepsilon_{\text{burn}}(\boldsymbol{\eta}, B)$ defined in Equation (58). Frozen-pool selection uses the full-sample ERM rule in Equation 21. Kernel smoothness, length scale, and the Ivanov radius are fixed once before the architecture sweeps and are not retuned separately for each feature dimension; every method uses the same predeclared RMS-distance rule $\|z - z'\|_2 / \sqrt{p}$ for its own feature dimension p .

C.2 The 36-task Legendre atlas

Every atlas target is deterministic and bounded by one, so $Y_t = F^*(X_{-\infty:t})$ and $\mathcal{R}^* = 0$. Coefficients are normalized by the sum of their absolute values as in Equation (54). The atlas contains six families.

Memory-depth family: eight tasks. For every $L \in \mathcal{G}_L$,

$$F_L^{\text{mem}}(\mathbf{x}^-) := P_2(x_{-L,1}). \quad (88)$$

Only the oldest relevant lag changes.

Multiscale-mode family: four tasks. For $H \in \mathcal{G}_H$, define

$$F_H^{\text{multi}}(\mathbf{x}^-) := \frac{1}{H} \sum_{\ell \in \mathcal{L}_H} P_2(x_{-\ell,1}), \quad (89)$$

with

$$\mathcal{L}_1 = \{8\}, \quad \mathcal{L}_2 = \{1, 64\}, \quad \mathcal{L}_4 = \{1, 4, 16, 64\}, \quad \mathcal{L}_8 = \{0, 1, 2, 4, 8, 16, 32, 64\}.$$

Thus increasing H increases the number of simultaneously required temporal modes while keeping degree two and one active coordinate per mode.

Spatial-load family: seven tasks. For $d \in \mathcal{G}_d$,

$$F_d^{\text{sp}}(\mathbf{x}^-) := \frac{1}{d} \sum_{j=1}^d P_2(x_{0,j}). \quad (90)$$

Every ambient coordinate is label-relevant, so increasing d cannot be made trivial by placing the target in one fixed low-dimensional subspace.

Nonlinear-degree family: six tasks. For $p \in \mathcal{G}_p$,

$$F_p^{\text{deg}}(\mathbf{x}^-) := P_p(x_{-4,1}). \quad (91)$$

The active lag and arity remain fixed while polynomial order changes.

Interaction-arity family: five tasks. Fix ambient dimension $d = 16$ and lag four. For $\iota \in \mathcal{G}_\iota$, define

$$F_\iota^{\text{int}}(\mathbf{x}^-) := \prod_{j=1}^{\iota} P_{16/\iota}(x_{-4,j}). \quad (92)$$

Because every ι divides 16, all five targets have total polynomial degree sixteen. The ambient dimension and memory lag are fixed; what changes is how that degree is distributed over distinct active coordinate factors.

Composite family: six tasks. The final cases combine difficulty axes rather than changing only one:

$$F^{\text{C1}}(\mathbf{x}^-) := P_8(x_{-64,1}), \quad \text{long-memory nonlinear,} \quad (93)$$

$$F^{\text{C2}}(\mathbf{x}^-) := P_{16}(x_{-32,1}), \quad \text{long-memory high-order,} \quad (94)$$

$$F^{\text{C3}}(\mathbf{x}^-) := \frac{1}{16} \sum_{j=1}^{16} P_4(x_{0,j}), \quad \text{current high-spatial nonlinear,} \quad (95)$$

$$F^{\text{C4}}(\mathbf{x}^-) := \frac{1}{16} \sum_{j=1}^{16} P_2(x_{-32,j}), \quad \text{long-memory high-spatial,} \quad (96)$$

$$F^{\text{C5}}(\mathbf{x}^-) := \frac{1}{8} \sum_{q=1}^8 P_2(x_{-\ell_q,q}), \quad (\ell_q)_{q=1}^8 = (0, 1, 2, 4, 8, 16, 32, 64), \quad \text{multiscale spatial,} \quad (97)$$

$$F^{\text{C6}}(\mathbf{x}^-) := \frac{1}{4} \sum_{q=1}^4 P_2(x_{-\ell_q,2q-1}) P_2(x_{-\ell_q,2q}), \quad (\ell_q)_{q=1}^4 = (1, 4, 16, 64), \quad \text{multiscale interactions.} \quad (98)$$

Together the six families contain $8 + 4 + 7 + 6 + 5 + 6 = 36$ targets. The one-axis families identify mechanisms; the composites test whether the same resource ordering survives when several sources of difficulty coexist.

C.3 Mixer-mechanism controls

The mixer ablation in Campaign I is deliberately narrower than a generic circuit search. For every paired random seed we evaluate exactly three versions of the same branch: the identity mixer $W_n = I$, the sampled single-qubit layer only $W_n = V$, and the full cycle mixer $W_n = E_{\tau(J_n)} \cdots E_{\tau(1)} V$. The latter two reuse the same sampled single-qubit axes and angles; the full mixer adds the sampled edge gates. No variant receives a separately tuned readout or memory-rate budget.

Each mixer is evaluated with both the vertex-only bank $\mathcal{O}_V(n)$ and the complete bank $\mathcal{O}_n$. We use the fixed mechanism architecture $n = 8$, $R = 8$, and $\tau_+ = 8$; no claim depends on this being an optimal architecture. For each paired repetition, first draw and freeze the shared $\mathbf{\Pi}_n$, write $\alpha_j(x) = \vartheta((\mathbf{\Pi}_n x)_j)$, and then generate

the three bounded current-time targets

$$F_1^{\text{mix}}(\mathbf{x}^-) := \cos \alpha_1(x_0), \quad (99)$$

$$F_2^{\text{mix}}(\mathbf{x}^-) := \sin \alpha_1(x_0) \sin \alpha_2(x_0), \quad (100)$$

$$F_3^{\text{mix}}(\mathbf{x}^-) := \cos \alpha_1(x_0) \sin \alpha_2(x_0) \sin \alpha_3(x_0). \quad (101)$$

All six mixer/bank variants share that preprocessor and label sequence. The first target is a one-site control; the second is directly compatible with an identity edge observable and with the mixed vertex term in corollary 3; the third matches the mixed edge-seed three-stream term. Defining these targets after the independent preprocessor draw is intentional: they test the theorem in its natural projected coordinates without letting Gaussian preprocessing itself confound the interaction-order ablation. They are therefore reported only as mechanism diagnostics and are kept separate from the preprocessor-independent 36-task atlas.

To test whether the same mechanism matters beyond targets tailored to the encoding coordinates, we repeat the identical six mixer/bank variants on the three fixed atlas tasks

$$F_8^{\text{mem}}, \quad F_{16}^{\text{int}}, \quad F^{\text{C6}}.$$

These labels are generated independently of the Gaussian preprocessor. The first is a low-interaction negative control, while the latter two require higher-order spatial or spatiotemporal interactions. No mixer-specific hyperparameter is changed between the theorem-aligned and atlas-aligned slices.

The complete and vertex-only observable banks also receive one finite-shot cost-matched comparison using the full cycle mixer on F^{C6} . A dyadic value M_E^{cmp} is frozen in the pre-run manifest. The complete bank uses all nine global Pauli settings at M_E^{cmp} repetitions per setting. The vertex-only bank requires only the three all- X , all- Y , and all- Z settings and uses $M_V^{\text{cmp}} = 3M_E^{\text{cmp}}$ repetitions per setting. Thus both variants consume exactly $9M_E^{\text{cmp}}$ branch-circuit runs per input window. Training and held-out measurement caches remain independent exactly as in Campaign II.

For exact circuit diagnostics, expand $W_n^\dagger O W_n = \sum_P c_{O,P} P$ in the normalized Pauli basis and report

$$\text{Spr}(W_n; \mathcal{O}) := \frac{1}{|\mathcal{O}|} \sum_{O \in \mathcal{O}} \sum_{\text{supp}(P) \not\subseteq \text{supp}(O)} |c_{O,P}|^2. \quad (102)$$

The score is identically zero for $W_n = I$ and for the local-only V layer, whereas the graph-local spreading result predicts positive mass outside the seed support for the relevant full-mixer observables almost surely. We also record the stable rank of the column-centered exact feature matrix on a common input sample, using exactly $r_{\text{stab}}(F_c) = \|F_c\|_F^2 / \|F_c\|_2^2$ with $r_{\text{stab}}(0) = 0$, as defined in Equation 70. These diagnostics test the proposed mechanism—operator spreading and observable enrichment—without treating performance alone as proof that entanglement is useful.

C.4 Maximal-cache construction and paired randomness

The large grids are made economical by coupling conditions without changing any required marginal distribution. The maximal caches are campaign-local: R_{max} for the architecture study, M_{max} for the $S = 1$ measurement study, and $(S_{\text{max}}, R_{\text{search, max}})$ for the exact-feature search study. They are never multiplied into one unnecessary maximal Cartesian cache.

Reservoir prefixes. For every Campaign I design, draw $R_{\text{max}} = 64$ independent branch seeds (U_r, ω_r) with $U_r \sim \text{Unif}(0, 1)$. At memory endpoint λ_+ set

$$\lambda_r(\lambda_+) := \lambda_0 \left(\frac{\lambda_+}{\lambda_0} \right)^{U_r}. \quad (103)$$

For each fixed λ_+ this has exactly the density in Equation (12), while reuse of the same U_r gives a paired comparison across memory endpoints. The architecture with R branches uses only the first R pairs. Consequently all values in $\mathcal{G}_R$ are available from one R_{max} draw.

Frozen-pool prefixes. Generate $S_{\max} = 256$ candidate designs once for Campaign III, each with $R_{\text{search},\max} = 32$ exact-feature branches. The setting with pool size S and branch count R uses candidates $1:S$ and branches $1:R$, then applies the identical full-sample ERM selection rule. Oracle-best curves are computed on the same prefixes. This preserves the order-statistic nesting of random search, yields every $S \in \{1, 2, 4, 8, 16, 32, 64, 128, 256\}$ and $R \in \{1, 2, 4, 8, 16, 32\}$ from one maximal exact-feature pool, and removes irrelevant between-pool Monte Carlo noise.

The compact selection-pressure slice uses the same $S_{\max} = 256$ candidate ordering at fixed $(n, R, \tau_+) = (8, 16, 32)$, with $N \in \{256, 1024, 4096\}$ and $S \in \{1, 16, 64, 256\}$. Within a paired repetition, the $N = 256$ and $N = 1024$ training sets are prefixes of the $N = 4096$ gapped training realization, so increasing N changes only the amount of observed data. Held-out evaluation uses one common independent test realization across all (N, S) cells. The slice is run only on F_g^{multi} and F^{C6} and uses exact features, so it isolates finite-pool selection from measurement noise.

Measurement prefixes. For Campaign II, where $S = 1$, generate $M_{\max} = 8192$ independent outcomes for every input window, tested branch, and one of the nine global Pauli settings. The feature estimate at repetition count M uses outcomes $1:M$. Since a single global outcome supplies all compatible Pauli coordinates, this cache preserves the within-shot correlations of the grouped measurement scheme. The total acquisition required to support an entire M sweep at a fixed tested architecture is therefore the maximal cache once, not $\sum_{M \in \mathcal{G}_M} 9NMR$ separate acquisitions. The representative finite-shot pool-ranking check in Campaign III is generated separately only up to $S = 64$ at its chosen dyadic M ; it does not instantiate the full $S_{\max}M_{\max}$ product.

Coupled Gaussian preprocessors. For a task of ambient dimension d , draw one matrix $G \in \mathbb{R}^{n_{\max} \times d}$ with $n_{\max} = 12$ and i.i.d. standard-normal entries. For $n \in \mathcal{G}_n$, set

$$\mathbf{\Pi}_n := \frac{1}{\sqrt{n}} G_{1:n, :} \quad (104)$$

Each $\mathbf{\Pi}_n$ has the exact marginal law in Equation (8), while widths share as much preprocessing randomness as possible. The random mixing circuit itself is still sampled from the correct width-specific law $\mathbb{P}_n^\omega$; width comparisons are paired by task, temporal sample, preprocessor seed, and random-seed stream rather than interpreted as nested quantum circuits.

Finite-window causal-proxy construction. For the window-fidelity experiment, the same temporal trajectory is propagated through each branch after a burn-in $B = 1024$. The resulting state is a *certified causal proxy*, not an exact infinite-past state. Contraction bounds its branchwise trace-distance residual by $2\lambda_+^B$, and the same feature/readout argument as in the finite-window theorem bounds the associated risk discrepancy by $\varepsilon_{\text{burn}}(\boldsymbol{\eta}, B)$. At the largest planned $\tau_+ = 64$ this branch-state residual is $2e^{-16} \approx 2.25 \times 10^{-7}$. The finite- w state is obtained by resetting to ρ_+ exactly w steps before the same label time and replaying the identical suffix. Every reported window gap is accompanied by $\varepsilon_{\text{burn}}(\boldsymbol{\eta}, B)$ so numerical burn-in error cannot be mistaken for finite-window error.

C.5 Homogeneous-memory control

The heterogeneous-memory comparison must preserve branch count and frozen mixing diversity. For a given R , retain the same $\omega_1, \dots, \omega_R$ but replace the independently drawn rates by one common rate. The primary control uses the geometric center of the log-rate interval,

$$\lambda_{\text{hom}} := \sqrt{\lambda_0 \lambda_+}, \quad (105)$$

so no extra tuning budget is introduced. On the representative multiscale slices used for the architectural claim, we additionally evaluate $\tau_{\text{hom}} \in \mathcal{G}_{\tau_+}$ and choose the common rate by the same full-sample empirical-risk rule. A heterogeneous multiplex must therefore be competitive not only with one fixed common memory scale but also with a full-sample-ERM-selected homogeneous alternative.

C.6 Baselines and matched resources

Projection-only Matérn control. This control is used only in the width-versus-spatial-load study. For the same shared Gaussian matrix used by QuaRK at width n , it receives

$$(\Pi_n x_{-w_{\text{ref}}+1}, \dots, \Pi_n x_0) \in \mathbb{R}^{nw_{\text{ref}}},$$

flattens the projected window in time order, and fits the same fixed RMS-normalized Matérn/Ivanov readout. It has no reservoir dynamics, mixer, or observable bank. Its purpose is not to compete as a third general baseline, but to expose empirically how much predictive information survives Π_n before the quantum representation acts. It therefore uses the same projection realization, temporal sample, kernel parameters, Ivanov radius, and held-out windows as the corresponding QuaRK width condition.

Direct finite-window Matérn. The raw baseline receives the same observed history $x_{-w_{\text{ref}}+1:0}$, flattened in time and coordinate order, and uses the same normalized Matérn kernel family, the same RMS-distance rule $\|z - z'\|_2 / \sqrt{p}$ for its own feature dimension p , and the same fixed Ivanov policy. It tests whether the controlled target requires a reservoir representation at all.

Classical multiscale reservoir. The classical reservoir consists of R independent frozen leaky branches driven by the same input and concatenated before the same Matérn readout. Its forgetting parameters are matched branchwise to the QuaRK λ_r values, and its total exposed state dimension is matched to $m_\eta = 12Rn$ up to integer allocation across branches. Input and recurrent weights are frozen random draws; no internal parameter is trained. The same RMS-distance rule is applied to its concatenated feature vector, so changing exposed dimension does not trigger a separate kernel retuning. The comparison therefore preserves the two structural ingredients most relevant to the QuaRK claim—parallel frozen dynamics and heterogeneous memory—while changing the quantum feature generator.

Neither baseline is intended to establish a quantum-advantage theorem. The raw kernel can reveal that a reservoir is unnecessary, and the classical multiscale reservoir can reveal that any observed gain follows from generic multiscale reservoir structure rather than specifically quantum features.

C.7 Diagnostics and paired reporting

Every resource curve uses paired temporal samples and paired design seeds where the construction above permits it. Except for the training-risk side of $\hat{\Delta}_{\text{gen}}$, all reported predictive risks are evaluated on held-out windows; paired quantities use the same held-out windows and labels, and any measured evaluation uses measurement randomness independent of the training cache. The final numerical study should use at least sixteen paired repetitions for architecture and fidelity curves and thirty-two for the S -selection curves, where order-statistic variability is larger. A long independent test realization is used for all population-risk approximations.

The primary metric is Equation (59). Campaign I’s width study reports the raw-history Matérn baseline, projection-only Matérn control, and QuaRK on the same paired width conditions. Its mixer mechanism slice additionally reports Spr from Equation (102) and exact-feature stable rank for the identity, local-only, and full-cycle mixers under vertex-only and complete observable banks. The same mixer/bank comparison is reported separately on the three preprocessor-independent atlas controls, and the finite-shot vertex/full-bank check is compared at identical branch-circuit run count. Campaign II reports $\hat{\Delta}_{\text{shot}}$, $\hat{\Delta}_{\text{win}}$, and $\hat{\Delta}_{\text{gen}}$, always evaluating the same fitted readout on the two feature/risk objects being compared. Campaign III reports ERM-selected NEMSE, oracle-best-in-prefix NEMSE, selection regret, and $C_Q = 9NMSR$. Its compact $N \times S$ slice reports the same oracle and regret quantities as a surface or grouped curves, so the effect of additional data on ERM over-selection is visible directly. The observed regret is the empirical check on the finite-sample selection price that is controlled theoretically by the uniform $\log S$ term; the certificate terms are not tabulated merely because they exist.

Finally, because the feature dimension $12Rn$ changes with architecture, the release records the empirical distribution of pairwise Matérn similarities computed with the predeclared RMS metric at each (n, R) setting.

This is a sanity diagnostic for residual kernel saturation, not another tuned hyperparameter. The kernel specification is not changed after inspecting the diagnostic: any residual collapse toward uniformly zero or uniformly one similarity is reported as a limitation of that fixed readout specification, not repaired by architecture-specific length-scale tuning.

The final empirical release will contain the exact generator implementation, all task identifiers from Sections C.2 and C.3, temporal and quantum random seeds, maximal branch/pool/measurement caches, baseline implementations, resource configurations, train/test indices, and the scripts that map those objects to every reported figure and table.